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Label-free metabolic imaging monitors the fitness of chimeric antigen receptor T cells

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Abstract

Chimeric antigen receptor (CAR) T cell therapy for solid tumors is challenging because of the immunosuppressive tumor microenvironment and a complex manufacturing process. Cellular

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Authors contributions: These authors contributed equally: D.L.P, D.C. D.L.P and C.W performed OMI imaging and analysis of T cell samples, and performed T cell isolation and activation. D.C, AT, MB isolated and cultured T cells and generated electroporation-based CRISPR-edited CAR T cells. K.P.M performed viral transductions. M.H.F performed mouse study. J.V, A.L, and K.S performed ATP production and cellular reducing potential analysis on activated T cells. J. L. performed the metabolomics analysis of two media and the IncuCyte analysis. D.L.P, D.C, M.H.F, K.P.M, C.W, A.T, J.V, A.L, K. Sylvester, J.L, and M.B performed experiments and analyzed the data. D.L.P and M.C.S* wrote the manuscript with inputs from all authors. C.M.C, J. F., K.Saha*, and M.C.S* supervised the research.

Competing interests

KS receives honoraria for advisory board membership for Andson Biotech and Notch Therapeutics. CMC receives honoraria for advisory board membership for Bayer, Elephas Biosciences, Nektar Therapeutics, Novartis, and WiCell Research Institute. MCS, DP, KS, DC declare two pending patents based on this work (WO2024229098A3, WO2024229099A1). MCS is an advisory board member for Elephas Biosciences. No other conflicts of interest are reported.

manufacturing protocols directly impact CAR T cell yield, phenotype, and metabolism, which correlates with in vivo potency and persistence. Though metabolic fitness is a critical quality attribute, how T cell metabolic requirements vary throughout the manufacturing process remains unexplored. Here, we use optical metabolic imaging (OMI), a non-invasive, label-free method to evaluate single-cell metabolism. Using OMI, we identified the impacts of media composition on CAR T cell metabolism, activation strength and kinetics, and phenotype. We demonstrate that OMI parameters can indicate cell cycle stage and optimal gene transfer conditions for both viral transduction and electroporation-based CRISPR/Cas9. In a CRISPR-edited anti-GD2 CAR T cell model, OMI measurements allow accurate prediction of an oxidative metabolic phenotype that yields higher in vivo potency against neuroblastoma. Our data supports OMI as a robust, sensitive analytical tool to optimize manufacturing conditions and monitor cell metabolism for increased CAR T cell yield and metabolic fitness.

Chimeric antigen receptor (CAR) T cells are genetically engineered to target tumor-associated antigens and perform cytotoxic functions to eliminate cancer cells. CAR T cell therapy has changed the approach and outlook for cancer care across several hematological malignancies, with patients from early trials maintaining decade-long remission ⁽¹⁾. Despite the anticipation about its application to solid tumors, the translation of CAR T cell therapy has yet to succeed ⁽²⁾, in part because the successful clinical translation of CAR T cells for solid tumors involves a complex manufacturing process, each step of which requires optimization to enhance CAR T functions against the immunosuppressive tumor and tumor microenvironment. The general CAR T manufacturing workflow includes activation of T cells isolated from leukapheresis products from patients with cancer using antibody stimulation, introduction of the CAR transgene, and expansion of the resultant CAR T cells to reach sufficient dosage ⁽³⁾.

First, antibody activation is critical to prime T cells for proliferation and genome editing. Though the effects of certain culture media and stimulating antibodies on T cell expansion and phenotype have been characterized ^(4, 5), their synergistic impacts on CAR T cell metabolism, functions, and transgene incorporation efficiency have yet to be explored. This knowledge gap makes the selection for optimal activation conditions challenging. Second, while viral transduction is the common CAR gene transfer method used in all six CAR T products approved by the United States Food and Drug Administration, it suffers from batch-to-batch variability ^(6,7) and safety concerns due to random transgene insertion ⁽⁸⁾. Meanwhile, despite promising in vivo potency ^(9,10), CRISPR-edited CAR T cells face challenges such as low transgene incorporation efficiency and viability ⁽¹¹⁾. Extended ex vivo culture is thus needed to reach the desired CAR T dosage, but comes with the risk of inducing terminal differentiation that decreases potency ^(12,13) while delaying patient treatments. Considering remarkable metabolic changes throughout the cell cycle, T cell metabolic states following activation can potentially indicate cell cycle stage and, thus, the optimal CAR gene transfer timeframe ^(11,14,15). Identifying this optimal time for CAR gene transfer could maximize CAR transgene incorporation efficiency and, hence, minimize the expansion time to reach the desirable dosages. This not only limits T cell differentiation to enhance CAR T potency but also shortens treatment wait-time and increases patient access. Third, similar to media composition and stimulating antibodies used during activation,

media composition and cytokines used during expansion can impact CAR T cell phenotype, fitness, and clinical outcomes (¹⁶). Several cytokine cocktails, including IL-7 and IL-15, have been studied to control CAR T cell differentiation and retain stem-like characteristics (e.g., expression of CCR7, CD62L, CR45RA) for in vivo potency and persistence (^{17,18,19}). Similarly, metabolites in culture media such as glucose and glutamine could condition CAR T cell metabolism and program their epigenetics for enhanced in vivo potency and persistence (16, 18), though their specific contributions at different stages throughout manufacturing have yet to be characterized.

T cell metabolic needs vary throughout manufacturing due to distinct functional demands at each stage. For instance, during activation, metabolic reprogramming is necessary to initiate cell cycle entry and enhance homology-directed repair efficiency for transgene incorporation. Meanwhile, during expansion, fine-tuning T cell metabolism is needed to support proliferation while minimizing terminal differentiation and exhaustion. Several studies suggest that metabolic conditioning of T cells during manufacturing enables in vivo adaptation to the immunosuppressive tumor microenvironment, which is characterized by high metabolic stress, nutrient scarcity, and toxic metabolic waste (^{20,21}). Metabolic fitness – as indicated by features like high dependence on oxidative phosphorylation and fatty acid oxidation, low glycolytic activity, and high mitochondrial mass – has recently emerged as a key critical quality attribute for CAR T cells (^{22–25}). In sum, metabolism plays an intricate role in T cell activation, cell cycle progression, gene transfer efficiency, and in vivo potency. Thus, monitoring T cell metabolism during CAR T manufacturing could enable fine-tuning of manufacturing conditions to match the dynamic metabolic demands of T cells at different phases.

However, the integration of standard metabolic assays into a real-time, non-invasive, adaptive CAR T manufacturing workflow remains challenging. These methods often lack single-cell resolution to characterize heterogeneity within the bulk T cell product (e.g., metabolite quantification from media or extracellular flux analysis), involve manipulation of the CAR T product, or are time consuming (e.g., single-cell metabolomics or flow cytometry). By contrast, optical metabolic imaging (OMI) is a label-free, noninvasive method to characterize metabolism within single cells (^{26,27}). OMI includes 13 metabolic parameters based on autofluorescence intensities and lifetimes of NAD(P)H and FAD, two metabolic coenzymes that are involved in hundreds of reactions in the mitochondria and cytosol, as well as cytoplasm size (Table 1). Since reduced NAD(P)H and oxidized FAD are autofluorescent, the optical redox ratio (Table 1), can be used to measure cellular redox balance (^{28–30}). Meanwhile, fluorescence lifetimes indicate the binding activity of these molecules. Due to conformational changes upon protein binding, free NAD(P)H self-quenches and has a short lifetime, while protein-bound NAD(P)H has an extended conformation and hence, longer lifetime. FAD displays an opposite trend, with free and protein-bound FAD having a long and short lifetime, respectively (^{29–31}). OMI provides single-cell resolution to characterize heterogeneity within population and identify important cell subsets while offering non-invasive, label-free, and real-time readouts of cell metabolism (26, 32).

Here, we test whether OMI measurements can inform manufacturing conditions that improve gene transfer efficiency and metabolic fitness of CRISPR-edited anti-GD2 CAR T cells for neuroblastoma treatment. Neuroblastoma is the most common extracranial solid tumor in children with a five-year survival rate of about 60% in high-risk patients (^{33,34}). Anti-GD2 CAR T cells are a promising treatment due to ubiquitous overexpression of the disialoganglioside GD2 on neuroblastoma cells (^{34,35}); however, poor T cell persistence and potency remains a substantial roadblock to clinical success. Using OMI, we characterized metabolic changes in T cells upon activation in various conditions and determined the relationship between T cell metabolism, cell cycle progression, and CAR gene transfer efficiency. We also performed OMI to determine the synergistic impacts of culture media and cytokine cocktails on CAR T cell metabolism and phenotype. Finally, OMI parameters were used to predict the expansion condition that achieved high in vivo potency in a NOD/SCID/IL2R γ c^{-/-} (NSG) mouse xenograft model of human neuroblastoma. Overall, our results demonstrate and validate a method to improve CAR T cell yield and efficacy, while highlighting the implication of T cell metabolism for successful CAR T clinical translation.

RESULTS

OMI shows media impacts on T cell metabolism and activation

Several media formulations with different concentrations of key nutrients such as glucose and glutamine are used for CAR T production in preclinical and clinical settings (^{36,37}). Among the most common are ImmunoCult XF media and TexMACS media (³⁶). Liquid chromatography-mass spectrometry analysis revealed these two T cell media had significantly different composition, with ImmunoCult XF media having higher glucose and glutamine levels (glucose⁺⁺/glutamine⁺), while TexMACs media relies on GlutaMax as the source for glutamine and has lower glucose levels (glucose⁺/glutamine⁻) (Extended Data Fig 1A–C). Higher pyruvate and arginine concentrations were also detected in TexMACs media, both of which have also been shown to regulate T cell metabolism and functions, in addition to glucose and glutamine (Extended Data Fig 1D–E). Besides, various soluble T cell activation antibodies have also been developed with different structures and compositions to control activation efficiency, such as StemCell α CD2/ α CD3/ α CD28 and TransAct α CD3/ α CD28. Using OMI, we characterized changes in T cell metabolism under various activation conditions to understand the impact of media composition and activator choice on the first step of the current CAR T manufacturing process. T cells were activated by four unique combinations of the two antibodies and two media above for 24, 48, and 72 hours and imaged with OMI (Fig. 1A, B). Quiescent T cells cultured in ImmunoCult XF or TexMACS media were also imaged at 24, 48, and 72 hours as the control for activation effect. At each time point, activated T cells showed lower NAD(P)H mean lifetime (NAD(P)H τ_m), higher proportion of free NAD(P)H (NAD(P)H α_1), and increased cell size compared to quiescent cells (Fig. 1B, C). High NAD(P)H α_1 and low NAD(P)H τ_m have been correlated with increased glycolysis (²⁶), suggesting that T cells undergo metabolic shift towards glycolysis upon activation, consistent with prior metabolic studies (³⁸). Consistently across 3 donors, T cells activated in ImmunoCult XF media displayed peak increase in the NAD(P)H α_1 at 48 hours post activation for both activating antibodies used (Fig. 1C, right). Meanwhile, T cells activated in TexMACS media demonstrated peak

increase in NAD(P)H α_1 at 72 hours post activation (Fig. 1D, right). This suggests that media composition, not activating antibodies, determines the kinetics of metabolic changes in activated T cells.

As both StemCell and TransAct antibodies were soluble, and several CAR T manufacturing processes currently rely on Dynabead activation, we further assessed the impact of culture media on T cell activation and metabolism under α CD3/ α CD28 Dynabead activation conditions (Extended Data Fig. 1). T cells from healthy donors were activated at different bead-to-T cell ratios to induce various stimulation strength in either ImmunoCult XF or TexMACs media. OMI characterization revealed a peak in NAD(P)H α_1 at 48 hours post activation in T cells activated in ImmunoCult XF media with both 1:1 and 2:1 bead-to-T cell ratios (Extended Data Fig. 1G). Meanwhile, T cells activated with α CD3/ α CD28 beads in TexMACs media exhibited a significant increase in NAD(P)H α_1 at 72 hours compared to at 24 hours (Extended Data Fig. 1H). These metabolic kinetics were consistent with those observed for T cells activated with TransAct and StemCell soluble antibodies in ImmunoCult and TexMACs media. Overall, our data indicated that the effect of media composition on T cell metabolic kinetics upon activation was conserved across several stimulation methods (Dynabead vs soluble antibodies) and strengths (1:1 vs 2:1 bead-to-T cell ratio).

Metabolic reprogramming upon activation is crucial to balance the energy demands and biosynthesis that supports T cell proliferation and effector functions (³⁹). Thus, we assessed ATP production and cellular reducing potential of activated T cells via bioluminescence-based assay to determine the impact of activation conditions on T cell function and their readiness to proliferate. Early after activation, T cells in glucose⁺⁺/glutamine⁺ ImmunoCult XF media exhibited significantly higher intracellular ATP levels, peaking at 48 hours, compared to those in glucose⁺/glutamine⁻ TexMACS media. However, this trend reversed at 72 hours post activation, when T cells activated in TexMACS reached peak ATP production (Fig. 1E). These changes in ATP production are consistent with the NAD(P)H α_1 kinetics as measured by OMI (Fig. 1D). We also measured cellular reducing capacity, an indicator of the overall bioreactivity of intracellular NAD(P)H-dependent reductase enzymes, to further investigate NAD(P)⁺ reducing metabolic pathways (such as glycolysis, pyruvate conversion to acetyl-CoA, or the tricarboxylic acid (TCA) cycle) (⁴⁰). T cells activated in ImmunoCult XF media displayed a rapid increase in cellular reducing capacity, plateauing at 72 hours, while T cells activated in TexMACS media exhibited a slower and continuous rise in reducing potential up to 72 hours (Fig. 1F).

Media composition not only impacts the kinetics of metabolic changes, but also influences the metabolic landscape of activated T cells overall. Uniform Manifold Approximation and Projection (UMAP) of OMI parameters showed clustering based on culture media rather than the type of activating antibodies used (Fig. 1G, H), indicating that metabolic features of activated T cells were predominantly determined by media composition. Distinct OMI metabolic profiles also allowed classification of T cells activated in ImmunoCult XF media and TexMACS media with high sensitivity and specificity (AUC > 0.93) across several classifier models (Fig. 1I). UMAP based on OMI measurements of T cells activated with Dynabead in ImmunoCult XF and TexMACs media showed similar clustering patterns, with

activated T cells clustered based on culture media rather than stimulating strength (bead-to-T cell ratios) (Extended Data Fig 1I–J). As there were several pairs of metabolic cofactors (NADH/NAD/NADPH/NADP) implicated in OMI measurements, we further assessed T cell metabolism upon activation from two other donors with independent bioassays. 72 hours post activation, we observed significantly lower NAD/NADH ratio, higher ATP levels, and higher LDH and GAPDH activity in T cells activated in ImmunoCult XF media compared to those in TexMACs media, consistently across several stimulation methods and strengths (Extended Data Fig 1K–O). Lower IDH activity was also observed in T cells activated in ImmunoCult XF compared to TexMACs media (Extended Data Fig 1P). The low NAD/NADH ratio, coupled with increased LDH and GAPDH activity, indicates a greater reliance on glycolysis, while reduced IDH activity suggests decreased TCA cycle engagement in T cells activated in ImmunoCult XF media compared to those activated in TexMACs. These findings align with our OMI data, highlighting the influence of media composition on T cell metabolism during activation.

Overall, our data demonstrate that while increased glycolysis upon activation is robust across several activation conditions, media composition controls the metabolic kinetics and ATP production of activated T cells, with faster changes induced by glucose⁺⁺/glutamine⁺ ImmunoCult XF media across various stimulation methods (soluble antibody vs Dynabead), composition (α CD2/ α CD3/ α CD28 vs α CD3/ α CD28), and strength (1:1, 2:1, and 3:1 bead-to-T cell ratio)

OMI indicates cell cycle progression in activated T cells

Since our data revealed the impact of media composition on T cell metabolism and energy production, both of which are directly involved in proliferation, we next investigated how different activation conditions affect cell cycle progression of activated T cells. Using OMI, we characterized T cell metabolism every 12 hours following two activation conditions: StemCell α CD2/ α CD3/ α CD28 antibody in ImmunoCult XF media (Imm) or TransAct α CD3/ α CD28 antibody in TexMACS media (Tex) (Fig 2A, Extended Data Fig. 2A). The signature decrease in NAD(P)H τ_m and increase NAD(P)H α_1 were detected in both Imm- and Tex-activated T cells as early as 12 hours post activation (Fig. 2B–D). However, consistent with the findings above, Imm-activated T cells reached peak metabolic changes (lowest NAD(P)H τ_m and highest NAD(P)H α_1) faster than Tex-activated T cells (36–48 hours compared to 48–72 hours post-activation, respectively) (Fig. 2C, D). We also observed greater effect sizes (Glass's delta) of the Imm activation condition on NAD(P)H τ_m and NAD(P)H α_1 compared to the Tex activation condition across three donors (Extended Data Table 1). These findings further support that glucose⁺⁺/glutamine⁺ ImmunoCult XF media induced not only faster but also more pronounced metabolic changes in activated T cells compared to TexMACS media. Besides metabolic changes, T cells also underwent a significant increase in cytoplasm size following activation by Imm and Tex methods, indicating that they are primed for proliferation (Fig. 2E).

Following activation with either Imm or Tex methods, T cells rapidly progressed through the cell cycle with increased DNA content as measured via flow cytometry of Hoechst stain (Extended Data Fig. 2B, C). Consistent with the metabolic kinetics observed with

OMI, Imm-activated T cells displayed an earlier increase in the percent of cells in S/G2/M phase (%S/G2/M) (36–60 hours) compared to Tex-activated T cells (60–72 hours) (Fig. 2F, G, Extended Data Fig. 2B, C). Correlation analysis confirmed the relationship between OMI metabolic features and cell cycle progression. Across three donors and two activation methods, we found significant ($p < 0.0001$) correlations between OMI measurements and %S/G2/M (Fig. 2H, I, Extended Data Fig. 2D). Specifically, %S/G2/M correlated negatively with NAD(P)H τ_m and positively with NAD(P)H α_1 and cytoplasm size, suggesting that actively cycling T cells are associated with low NAD(P)H τ_m , more free NAD(P)H, and large cytoplasm. T cell proliferation (%Ki-67⁺ cells) also trended upwards throughout the 72-hour activation time course (Extended Data Fig. 2E, F). In summary, using non-invasive, label-free OMI measurements, we have demonstrated the influence of media composition on cell cycle progression following activation, and the faster metabolic changes induced by glucose⁺⁺/glutamine⁺ ImmunoCult XF media support earlier cell cycle entry compared to glucose⁺/glutamine⁻ TexMACS.

OMI identifies T cell features that enhance CAR integration

Activation primes T cells for CAR gene transfer; however, our data indicate that different activation conditions yield different T cell metabolic profiles and cell cycle progression kinetics. Thus, we further investigated how OMI characteristics of T cells at the gene transfer timepoint affected transgene incorporation efficiency. Anti-GD2 CAR transgene was introduced using either retroviral transduction (Extended Data Fig. 3A–F) or electroporation-based CRISPR/Cas9 (Fig. 3). Briefly, T cells isolated from two healthy donors were activated (day 0) and underwent T-cell receptor alpha-constant knockout (*TRAC* KO) (day 2) prior to retroviral transduction with two anti-GD2 CAR constructs (14G2a-OX40-CD28- ζ CAR or 14G2a-41BB- ζ CAR) (day 4) as previously described (Extended Data Fig. 3A) (⁹). At the transduction timepoint (day 4), *TRAC* KO T cells displayed significantly lower NAD(P)H τ_m , higher NAD(P)H α_1 , and greater cytoplasm size – OMI features of cycling cells – compared to control T cells with intact T cell receptor (Extended Data Fig. 3B–E). Importantly, we observed significantly higher transduction efficiency, quantified as percent CAR positivity post-expansion (day 11), in *TRAC* KO cells compared to control cells (Extended Data Fig. 3F). This suggests that metabolic and morphological characteristics measured by OMI at the time of transduction can indicate the condition that increases transgene incorporation, and, hence, CAR yield.

Besides viral vectors, CRISPR/Cas9 is currently being explored for future CAR T therapies as it allows precise genome editing that improves treatment safety and efficacy (42, 43). Therefore, we next characterized whether OMI features could inform CRISPR genome editing efficiency. Following 12-to-72 hour activation by either Imm or Tex methods, T cells were imaged with OMI and electroporated to generate CRISPR-edited anti-GD2 CAR T cells as previously described (⁴³). Notably, across three donors, the timeframe that yielded optimal genome editing efficiency, quantified as percent CAR positivity post expansion (Fig. 3B, C), aligned with the timeframe of maximal decrease in NAD(P)H τ_m and increase in NAD(P)H α_1 (Fig. 2C, D). These OMI characteristics (low NAD(P)H τ_m and high NAD(P)H α_1) were consistent with those displayed by T cells that yielded high transduction efficiency (Extended Data Fig. 3A–F). For Imm-activated T cells, electroporation at 36–

60 hours post-activation yielded the highest CAR gene transfer efficiency (Fig. 3B). Meanwhile, Tex-activated T cells showed increased % CAR positivity as the activation duration at electroporation increased, with the highest genome editing efficiency observed for electroporation at 48–72 hours post-activation (Fig. 3C). Electroporation at the optimal timeframe yielded a 3- to 4-fold-increase in genome editing efficiency.

Overall, our data revealed a consistent relationship between OMI metabolic features, specifically low NAD(P)H τ_m and high NAD(P)H α_1 , and CAR transgene incorporation across both retroviral transduction and electroporation-based CRISPR/Cas9.

OMI features predict optimal timing for CAR gene editing

Unlike label-free non-invasive OMI, cell cycle analysis based on S/G2/M and Ki-67 measurements from flow cytometry are destructive to T cells and involve exogenous labelling of the culture. As we observed distinct OMI features in T cell groups with high versus low transgene incorporation efficiency, we sought to determine whether label-free OMI measurements could inform the electroporation timepoint decision and how its sensitivity and specificity compared to that of flow cytometry measurements. Correlation analysis across 36 samples from 3 donors revealed significant ($p < 0.05$, $|R| > 0.4$) correlations between genome editing efficiency (percent CAR positivity) post expansion and several OMI measurements at electroporation (Fig. 3D–F, Extended Data Fig. 3H, I). Hoechst median fluorescence intensity (MFI) of activated T cells at electroporation also significantly correlated with genome editing efficiency, though the R-value was slightly lower than those of OMI variables [$R = 0.42$ (Fig. 3G) compared to $R = -0.44$ and $R = 0.66$ (Fig. 3E–F)]. Gating based on Hoechst MFI yielded the percentage of cycling cells (%S/G2/M), which also correlated with %CAR positivity 7 days later (Extended Data Fig. 3J). Similarly, proliferative capacity (%Ki-67⁺) in T cells also showed significant ($p < 0.0001$) positive correlations with genome editing outcome (Extended Data Fig. 3K). These findings align with prior research showing that genome editing during mitosis increases transgene incorporation efficiency (11, 15). Additionally, our data indicate the potential of OMI to identify gene transfer conditions that maximize CAR T cell yield.

To further assess the accuracy of OMI for guiding the electroporation timepoint decision, we used machine learning to predict cells from samples that later exhibit high versus low editing efficiency (high efficiency is considered as >15% CAR⁺ for Imm-activated T cells and >10% CAR⁺ for Tex-activated T cells). A Random Forest classifier achieved good sensitivity and specificity (AUC = 0.83) in predicting genome editing efficiency (day 7) based on OMI parameters at the time of electroporation (day 0) (Fig. 3H). Meanwhile, models trained on Hoechst MFI to predict optimal electroporation conditions did not perform as well (AUC = 0.74) despite 150x more cells used for training (Fig. 3I). Overall, T cell metabolism and morphology, as quantified by non-invasive label-free OMI, can inform when activated T cells are optimal for CAR gene transfer to increase CAR yield.

OMI and phenotype analysis stratify CAR T cells by media

Ex vivo expansion following CAR gene transfer not only achieves sufficient CAR T dosage, but also shapes the metabolic and phenotypic features of the resultant CAR T

products, which highly correlate to potency and treatment outcomes. As OMI measurements have identified the important role of media composition on T cell metabolism during the activation and gene transfer phases of CAR T manufacturing, we further evaluated how two media compositions (ImmunoCult XF and TexMACS, as above) in combination with various cytokines (IL-2, IL-7, IL-7/IL-2, and IL-7/IL-15) control features of CRISPR-edited anti-GD2 CAR T cells. We used OMI and flow cytometry of 6 naïve/stem memory markers to evaluate CAR T cell metabolism and phenotypes under several expansion conditions. Both heatmap and UMAP of OMI measurements from CAR T cell products post-expansion (which included both CAR⁺ and CAR⁻ cells) revealed distinct clustering based on culture media rather than supplemented cytokines (Fig. 4B–D). Notably, CAR T cells expanded in TexMACS media exhibited lower FAD mean lifetime (FAD τ_m), regardless of cytokines used (Extended Data Fig. 4B). Machine learning algorithms achieved high sensitivity and specificity in classifying CAR T cells expanded in ImmunoCult XF media and TexMACS media based on their OMI features, further confirming their distinct metabolic profiles (AUC > 0.96) (Fig. 4E). Similar trends were observed in the CAR⁺ fraction with clustering based on culture media in a UMAP of OMI parameters (Extended Data Fig. 4C, D) and high accuracy for classification of CAR⁺ cells by media condition based on OMI features (Extended Data Fig. 4E). We did, however, observe consistently higher NAD(P)H τ_m in CAR T cells expanded in IL-7±IL-15 compared to those expanded in IL-2 in ImmunoCult XF media (Extended Data Fig. 4F), suggesting that CAR T cells did modulate their metabolism based on the cytokines present, though to a lesser extent than the culture media.

Importantly, media composition also had dominating impacts on CAR T cell phenotype (Fig. 4F–H). Flow cytometric analysis revealed significantly higher CCR7 expression and lower CD62L expression in CAR T cells expanded in TexMACS compared to those expanded in ImmunoCult XF media (Extended Data Fig. 4G, H). UMAP based on surface marker expression also showed distinct clusters of CAR T cells expanded in ImmunoCult XF versus TexMACS media, regardless of cytokines used (Fig. 4F, G, Extended Data Fig. 4I). Consistently, machine learning algorithms trained on surface marker expression to classify CAR T cells expanded in ImmunoCult XF versus TexMACS media also achieved high sensitivity and specificity (AUC = 0.97) (Fig. 4H). Interestingly, CAR T cells expanded in IL-2 showed a distinct phenotypic island that was not observed in other cytokine conditions (red circle, Fig. 4G). However, the impact of cytokines on CAR T cell phenotype was still secondary compared to culture media composition, shown by overlapping clusters of cytokine conditions in the OMI and surface marker UMAPs (Fig. 4D, G).

OMI shows distinct CAR T metabolic profile with media switch

Transient glucose or glutamine restriction during manufacturing has been shown to produce CAR T cells with higher metabolic fitness, characterized by greater mitochondrial mass and more oxidative activity, that correlates to better in vivo potency and persistence (18, 45, 46). As CAR T cells expanded in ImmunoCult XF versus TexMACS media exhibited different phenotypic and OMI profiles (Fig. 4), we further investigated how CAR T cell metabolic profile and function are impacted by changing the expansion condition from TexMACS + 10ng/mL IL-7 pre-electroporation, which served as a transient glucose and glutamine

restriction, to ImmunoCult XF + 500U/mL IL-2 post-electroporation (Fig. 5, Extended Data Fig. 5). Hereafter this condition is referred to as Tex→Imm.

Post-expansion and pre-infusion into a mouse xenograft model (Fig. 5A), we did not observe distinct T cell phenotypes in CAR T cells expanded in the Tex→Imm condition compared to those expanded ImmunoCult XF (Imm) and TexMACS (Tex) media alone. CAR T cells expanded in Tex→Imm displayed an intermediate phenotype based on naïve and stem memory markers CCR& and CD62L (Fig. 5B, Extended Data Fig. 5B). However, we observed significantly less lactate secreted by Tex→Imm CAR T cells, suggesting lower glycolytic activity, compared to CAR T cells cultured in a singular media composition (Fig. 5C). This observation was further supported by metabolic flux analysis that showed a significantly higher ratio of oxygen consumption rate (OCR) to extracellular acidification rate (ECAR) in non-edited T cells expanded in the Tex→Imm condition compared to those cultured in Tex media alone. The high OCR/ECAR ratio indicated an increased dependence on oxidative metabolic pathways in Tex→Imm group (Extended Data Fig. 5C).

Importantly, OMI also captured a distinct metabolic profile in CAR T cells expanded in Tex→Imm, which had significantly lower NAD(P)H α_1 and higher NAD(P)H τ_m than CAR T cells cultured in Imm or Tex media alone (Fig. 5E, F). As OMI allows single-cell measurements, we further investigated the population heterogeneity in CAR T cells expanded within the Imm, Tex, and Tex→Imm conditions. We observed two subpopulations in each CAR T cell group post-expansion, displaying high- and low-NAD(P)H τ_m respectively (Fig. 5G). Notably, the high NAD(P)H τ_m subpopulation made up a greater proportion in Tex→Imm CAR T cells (~30–40%) compared to the other two media conditions (10% in Imm and <5% in Tex CAR T cells). Additionally, independent OMI analysis of CCR7⁺ T cells from four donors showed significantly higher NAD(P)H τ_m and lower NAD(P)H α_1 than CCR7⁻ cells (Fig. 5H–J). This suggests a potential shift towards stem-like metabolism in Tex→Imm CAR T cells, which had more cells in the high-NAD(P)H τ_m subset (Fig. 5G, I). Overall, the distinct OMI metabolic profile of Tex→Imm CAR T cells correlates to a lower glycolytic rate and higher dependence on oxidative pathways that can potentially improve in vivo persistence and potency (22, 45).

To further explore the roles of glucose and glutamine in shaping the metabolic profile induced by media transitions, we supplemented TexMACs media with additional glucose (7.3mM) and glutamine (4.3mM) to match the concentrations found in ImmunoCult XF, creating a new condition referred to as Tex^{hi}. The specific amounts of glucose and glutamine added were determined through our LC-MS analyses. We activated and expanded T cells from two healthy donors under various conditions, including: ImmunoCult XF, TexMACs, Imm→Tex, Tex→Imm, Tex→Tex^{hi}. Media switch happened on day 2 for T cells activated in ImmunoCult XF media and day 3 for T cells activated in TexMACs media to represent the timepoint at which activating antibodies are removed and T cells undergo CAR gene transfer. All T cells were then expanded for 6 days and collected for OMI and other bioassays. Cell count throughout the expansion process indicated a substantial impact of expansion condition on T cell proliferation capacity. Specifically, Imm and Tex→Imm T cells demonstrated significantly higher expansion capacities compared to T cells expanded under Tex conditions (Extended Data Fig. 6A–B). This enhanced

proliferation was further supported by ATP production measurements, where Imm and Tex→Imm T cells produced significantly more ATP than other groups (Extended Data Fig. 6C). Notably, the media transition from TexMACs to ImmunoCult XF (Tex→Imm) resulted in intermediate NAD(P)H binding activity (as reflected by NAD(P)H τ_m) while Imm T cells displayed the lowest NAD(P)H τ_m (Extended Data Fig. 6D–F). Consistently, independent NAD/NADH measurements showed Tex→Imm and Imm T cells exhibited intermediate and the lowest NAD/NADH ratios, respectively (Extended Data Fig. 10G). Meanwhile, the low NAD(P)H τ_m and high NAD(P)H intensity in Imm T cells were consistent with high NADH concentration and elevated LDH and GAPDH activity to indicate high glycolytic activity in these cells compared to other groups (Extended Data Fig 6E–J).

Additionally, low NAD(P)H intensity in Tex→Imm and Tex→Texhi cells aligned with lower NADH concentration, (Extended Data Fig. 6F, H–J). Tex→Imm T cells also exhibited the lowest isocitrate dehydrogenase (IDH) activity, which is responsible for NAD(P)H production via the TCA cycle, further explaining their reduced NAD(P)H intensity (Extended Data Fig. 6K). Despite lower IDH activity, Tex→Imm T cells maintained intermediate ATP levels and NAD/NADH ratios, suggesting some degree of oxidative phosphorylation activity even with reduced TCA cycle flux. We also observed the lowest G6PDH and 6PGDH activity in Tex→Imm T cells, indicating reduced pentose phosphate pathway (PPP) contribution. This metabolic profile is distinct from Tex and Tex→Texhi T cells, which showed elevated NAD/NADH ratios and enhanced activity of PPP dehydrogenases (Extended Data Fig 6L, M). The heightened PPP activity in Tex and Tex→Tex^{hi} cells likely reflects a response to oxidative stress as evidenced by increased NAD/NADH ratio and a greater need for reactive oxygen species management. In contrast, Tex→Imm T cells, with their intermediate NAD/NADH ratio, seem to experience less oxidative stress. Overall, our data suggest that Tex→Imm cells possess low NADH levels, IDH and PPP activity, with intermediate glycolytic activity and sufficient ATP production.

CAR T cells with media switch yield better in vivo potency

As OMI and other metabolic assays revealed a distinct metabolic profile expressed by Tex→Imm CAR T cells, we further assessed how different expansion conditions influenced in vivo potency of CRISPR-edited anti-GD2 CAR T cells. NSG mice with established GD2⁺ CHLA-20-Luciferase neuroblastoma xenografts were treated with 4 million CAR⁺ T cells expanded under Imm, Tex, or Tex→Imm media conditions (Fig. 6A). By day 17 post treatment, CHLA-20 bioluminescence increased significantly in cohorts treated with CAR T cells expanded in Imm or Tex media alone, indicating tumor progression (Fig. 6B, C). Notably, CAR T cells expanded in Tex→Imm condition led to tumor regression (fold change in tumor flux < 1) in two out of four mice, and better control of tumor flux overall (Fig. 6B, C, Extended Data Fig. 7B, C). Interestingly, CD45⁺ human T cells isolated from the mouse spleen 21 days post treatment with Tex→Imm CAR T cells showed better retention of stem cell memory phenotype (CCR7⁺/CD62L⁺) and resistance to exhaustion in vivo (PD-1⁺/TIGIT⁺) compared to CAR T cells expanded by singular media (Fig. 6D, E, Extended Data Fig. 7D–G) (47, 48). This suggests that the distinct OMI metabolic profile observed in Tex→Imm CAR T cells ex vivo is correlated with functional advantages in vivo.

We hypothesized that the distinct metabolic profiles expressed by Tex→Imm CAR T cells pre-infusion underlie their persistence and efficacy in vivo. To test this, we investigated whether pre-infusion OMI metabolic features could be used to predict CAR T in vivo potency. A UMAP of OMI parameters (Table 1) from pre-infusion CAR T products revealed different clusters of Tex→Imm expanded CAR T cells versus CAR T cells expanded in singular media (Fig. 6F). Machine learning classification based on pre-infusion OMI metabolic profiles distinguished high potency (Tex→Imm) from low potency (Imm or Tex alone) CAR T cells with high sensitivity and specificity (Fig. 6G; AUC > 0.91). These findings align with previous research indicating better in vivo potency by stem cell memory T cells and suggest that OMI measurements of CAR T cells ex vivo can predict in vivo response (Fig. 6G).

To further explore the potential of media transitions in enhancing CAR T cell metabolic fitness and functionality, we assessed the cytotoxic capacity of Imm and Tex→Imm anti-GD2 CAR T cells across additional cancer models. Donor-matched Imm and Tex→Imm GD2 CAR T cells were imaged using OMI before co-culture with two additional GD2-expressing cancer cell lines: M21 melanoma and MG63 osteosarcoma. Both cancer cell lines expressed GFP. Anti-GD2 CAR T cells and cancer cells were cultured at four effector-to-target ratios (1:1, 1:2, 1:5, and 1:10) for 72 hours in a Sartorius IncuCyte instrument. Consistent with our previous findings, we observed significantly lower NAD(P)H α_1 and higher NAD(P)H τ_m in Tex→Imm CAR T cells compared to Imm CAR T cells (Extended Data Fig. 8A–C).

Cytotoxicity was measured by monitoring GFP intensity in both cancer cell lines, with lower GFP intensity or percentage of GFP+ cells over time indicating higher CAR T cell-mediated tumor cell killing (Extended Data Fig. 8D–K). Our results showed that throughout 72 hours both Tex→Imm CAR T cells and Imm CAR T cells demonstrated efficient in vitro cytotoxicity against GD2⁺ target cells (M21 and MG63). At 1:1 effector-to-target ratio, both CAR T cell groups performed similarly against M21 melanoma, while Imm CAR T cells exhibited higher cytotoxicity, reflected by lower normalized GFP intensity, compared to Tex→Imm CAR T cells in coculture with MG63. However, at lower effector-to-target ratios (1:5 and 1:10), Tex→Imm CAR T cells demonstrated enhanced cytotoxicity with lower GFP intensity and %GFP+ cells than Imm CAR T cells in both cancer models (Extended Data Fig 8D, E, H, I). Additionally, at 72 hours post co-culture, some Tex→Imm CAR T cell coculture groups, specifically 1:10 effector-to-target ratio with M21 melanoma and 1:2 effector-to-target ratio with MG63, showed significantly lower GFP intensity and %GFP+ cells compared to Imm CAR T cells at the same effector-to-target ratios (Extended Data Fig 8F, K). These results suggest that Tex→Imm CAR T cells may have enhanced performance under conditions of increased cancer cell burden, though this effect is dependent on the tumor model and the effector-to-target ratio.

DISCUSSION

The translation of CAR T cell therapies to solid tumors is challenging, partially due to a complex manufacturing process that requires optimization at several stages. Current methods to assess T cell function are invasive, involving handling and sampling an active cell culture,

and mainly rely on final product parameters to satisfy release criteria. Thus, they do not capture the dynamic and heterogeneous changes in CAR T cells throughout manufacturing, including metabolic changes. This limits the ability to fine-tune manufacturing conditions to achieve the optimal CAR T cell profile. Recent research suggests that T cell metabolism correlates with function, and metabolic reprogramming *ex vivo* can achieve desirable functions *in vivo* (22, 24, 45, 46). Here, we have demonstrated that OMI, a label-free optical imaging technique, can non-invasively monitor T cell metabolism during CAR T cell manufacturing and OMI features can be used to determine the optimal manufacturing conditions to improve CAR T cell products.

Our results highlight that nutrient availability and media composition modulate the strength and speed of T cell activation, while also controlling cell cycle entry and differentiation fate. We examined two common T cell culture media – ImmunoCult XF and TexMACS – which have different concentrations of key nutrients that impact T cell metabolism and functional fates. Specifically, ImmunoCult XF media has higher glucose and glutamine concentrations, while TexMACS media contains GlutaMax, which can be hydrolyzed into L-glutamine and L-alanine by cell-secreted aminopeptidases (⁴⁸). Interestingly, activation in ImmunoCult media induced faster and greater changes in T cell metabolism, ATP production, and cellular reducing potential. Similarly, expansion in ImmunoCult XF and TexMACS media also yielded CAR T cells with distinct metabolic and phenotypic profiles. In our study, the effects of media composition on T cell metabolism and function dominated over the effects of different activating antibodies and supplemented cytokines. These observations emphasize the importance of optimizing media formulations in the clinical development of CAR T cells.

Metabolic shifts during mitosis have been documented, with high aerobic glycolysis and glutaminolysis for energy production and biomass synthesis (^{49,50}). Our data reveal a correlation between T cell OMI features, notably high NAD(P)H α_1 and low NAD(P)H τ_m , and cell cycle stage analysis by flow cytometry of Hoechst and Ki-67 stains. Previously, high NAD(P)H α_1 and low NAD(P)H τ_m were associated with cell cycle entry in Kasumi-1 cells (human acute myeloid leukemia progenitors) (⁵¹). This suggests that OMI may be sensitive to metabolic changes due to the cell cycle across several human cells, and that label-free OMI measurements can be a good indicator of cell cycle progression in T cells following activation.

Successful incorporation of the CAR transgene is required for the expression of functional CAR receptors to mediate therapeutic efficacy. The efficiency of several gene transfer methods is controlled by cell cycle, as mitotic cells have high homology directed repair that enables precise, on-target genome editing (^{52–54}). Several studies have attempted cell cycle synchronization to improve CAR transgene incorporation efficiency (56, 57). Here, we have demonstrated a consistent relationship between OMI metabolic measurements and gene transfer efficiency across two methods (viral transduction and electroporation-based CRISPR/Cas-9). Machine learning models based on label-free OMI measurements accurately identified gene transfer conditions that later resulted in high CAR transgene incorporation efficiency, indicating that OMI could inform the timing of genome editing to improve CAR yield.

Ex vivo metabolic perturbation during manufacturing, such as transient glucose or glutamine restriction, have been shown to reprogram CAR T cells epigenetically and prime them for the metabolically stressed, immunosuppressive in vivo tumor microenvironment (58, 59). Here, we found that CAR T cells undergoing media switch from glucose⁺/glutamine⁻ TexMACS pre-electroporation to glucose^{high} glutamine^{high} ImmunoCult XF post-electroporation exhibited distinct OMI profiles, with high oxidative phosphorylation and low glycolytic activity, together with improved in vivo potency. This is consistent with previous studies on the relationship between glycolysis and oxidative phosphorylation metabolism and T cell exhaustion and persistence (^{59–61}), and suggests that modulating T cell metabolism based on different energy requirements at several manufacturing stages can improve CAR T efficacy. Interestingly, T cells expanded in the Tex→Texhi condition exhibited metabolic characteristics largely similar to those in the Tex condition, with some overlap with the Tex→Imm group, such as low NAD(P)H intensity and NADH concentration. These findings suggest that changes in glucose and glutamine levels during expansion had some influence on cell metabolism, but other components in the media, such as pyruvate and arginine, also played an important role in driving the metabolic shifts observed during the transition from TexMACs to ImmunoCult XF and should be further explored in future work.

We were able to predict CAR T cell groups with high versus low in vivo potency based on pre-infusion OMI metabolic features with high sensitivity and specificity. This indicates that OMI could screen and inform manufacturing conditions that yield high potency or predict treatment response based on ex vivo measurements pre-infusion. We also identify a substantial sub-population with higher NAD(P)H τ_m in CAR T cells expanded in Tex→Imm that potentially contributes to their increased in vivo potency and persistence. These observations emphasize the significance of OMI's single-cell resolution in characterizing population heterogeneity to identify therapeutically beneficial subsets.

There are important caveats to this study. T cells isolated from healthy donors were used as starting material for CAR T cell manufacturing, which does not fully capture characteristics observed in autologous T cells from cancer patients. Furthermore, the absence of endogenous TCR in our virus-free CRISPR *TRAC* KO anti-GD2 CAR T model may influence T cell response to media and cytokines. We also observed different metabolism between untransfected T cells and CAR T cells under various expansion conditions (Fig. 5 and Extended Data Fig. 6). Therefore, further research should explore the effects of media composition and cytokines on CAR T cells with an intact TCR and validate the relationship between OMI measurements and in vivo potency across various CAR T cell and tumor models. Other metabolic perturbations should be explored to further optimize CAR T cell metabolic fitness and functionality. While NAD(P)H and FAD are critical metabolic cofactors involved in key pathways such as glycolysis, the TCA cycle, and oxidative phosphorylation, they represent only a fraction of the cellular metabolic machinery. OMI provides valuable insights into the overall cellular metabolic state in single cells and, as a label-free method, offers substantial potential for inline monitoring during CAR T cell manufacturing. However, it is essential to complement OMI measurements with additional bioassays to provide a more comprehensive understanding of CAR T cell metabolism and functionality. This integrated approach will ensure accurate interpretation

and allow identification of key metabolic players that contribute to CAR T cell performance. The rapid 24-hour CAR T manufacturing process holds important potential to address several challenges faced by current practices, including lower T cell exhaustion and terminal differentiation while increasing supply capacity. Future research will focus on developing and validating OMI applications to optimize this promising process. Additionally, the current two-photon laser scanning OMI system has been adapted into a flow cytometer⁽⁶²⁾ or single-photon microscope^(63,64) to achieve faster speed, higher-throughput, automation, and a smaller footprint to better integrate into CAR T manufacturing workflows. These technical efforts will expand the throughput and scope of OMI as a non-invasive, sensitive tool to support the translation of CAR T cell therapy for solid tumors.

In summary we demonstrate that OMI is a sensitive and non-invasive approach for profiling T cell metabolism at multiple stages of CAR T cell manufacturing. Our results show consistent correlations between OMI parameters and key manufacturing outcomes, including gene transfer efficiency via retroviral vectors and electroporation-based CRISPR/Cas9, as well as metabolic phenotypes associated with cell cycle progression and expansion conditions. OMI also detected distinct metabolic profiles in Tex→Imm CAR T cells, which were associated with enhanced in vivo potency and, in certain contexts, improved in vitro cytotoxicity. We further validated OMI measurements against established metabolic assays, including metabolite quantification and extracellular flux analysis. These findings support the potential of OMI as a complementary analytical tool to help inform and optimize CAR T cell manufacturing, though further studies across different donors and tumor models are warranted⁽⁶⁵⁾.

MATERIALS AND METHODS

Study Design

Sample size for each in vitro experiment includes at least 3 donors with at least 50 cells/ experimental group/donor to capture intra- and inter-donor heterogeneity based on a previous power analysis for the sensitivity of OMI to T cell activation⁽²⁶⁾. For the mouse studies, two donors were used across two independent experiments and the number of mice per group was determined by prior study on the same virus-free CRISPR-edited anti-GD2 CAR T model (9), with at least 4 mice in each group receiving either media-transitioned (Tex→Imm) CAR T cells or non-media-transitioned (Imm or Tex) CAR T cells. Rules for stopping data collection was based on the 10-day CAR T cell manufacturing process set by prior protocols (9). Mouse tumors were followed until tumor volume reached 4000mm³. We did not exclude any data besides the mouse studies, in which one mouse was excluded due to out-of-range starting tumor size as determined by IVIS reading. We have not otherwise removed outliers. Primary and secondary endpoints were prospectively selected, and appropriate statistical corrections were applied to multiple endpoints. The number of repeats for each experiment is given in the figure caption to show that results were substantiated across multiple donors. Replication is performed at several levels: cell, image, and donor, with 4–7 technical replicates (images) per experiment. Each main finding of the paper is supported by 650–4,800 biologically independent T cells from two to three donors (findings include: OMI endpoints for classification and correlation analysis to determine

impacts of media composition on T cell activation, and to predict cell cycle entry, optimal gene transfer conditions, and CAR T metabolic fitness). Supportive experiments (ATP production, cellular reducing potential, extracellular flux analysis, lactate measurement) include at least 3 technical replicates from 1–2 donors.

The objectives of our research were to demonstrate that OMI can determine (1) cell cycle entry timeline, (2) optimal CAR gene transfer conditions, and (3) media compositions that enhance metabolic fitness of CAR T cells. These were pre-specified objectives. Research subjects were healthy volunteers and NSG mice bearing GD2+ Luciferase CHLA-20 tumors. This was a controlled laboratory experiment where treatments and measurement techniques are described below. Mice were randomly assigned among treatment groups as described below. The mouse studies were performed by M.H.F. and the investigator performing in vitro analyses of CAR T infusion products (D.L.P.) was blinded to the results of these mouse studies.

Measurement of media metabolites via LC-MS

To extract metabolites from culture media, 20µl of culture medium was added to 80µl of LC-MS grade methanol. Samples were then vortexed and centrifuged at 20,627g for 5 min at 4 C to remove any insoluble debris. Supernatant containing small molecules was further diluted 1:10 with LC-MS grade H₂O. Soluble metabolite samples were analyzed with a Thermo Q-Exactive mass spectrometer coupled to a Vanquish Horizon Ultra-High Performance Liquid Chromatograph. Metabolites were separated on a 2.1 × 100mm, 1.7 µm Acquity UPLC BEH C18 Column (Waters) with a gradient of solvent A (97:3 H₂O/methanol, 10 mM TBA, 9 mM acetate, pH 8.2) and solvent B (100% methanol). The gradient was: 0 min, 5% B; 2.5 min, 5% B; 17 min, 95% B; 21 min, 95% B; 21.5 min, 5% B. Flow rate was 0.2 ml min⁻¹. Data were collected with full scan. Identification of metabolites reported here was based on exact *m/z* and retention time that were determined with chemical standards. Data were collected with Xcalibur 4.0 software and analyzed with Maven. Concentration of metabolites were quantified from ion count in LC-MS analysis based on external calibration curves obtained by running chemical standards at various concentrations using the same LC-MS method.

T cell activation with different activating antibody and media

Peripheral blood was drawn from healthy donors under a protocol approved by the Institutional Review Board at the University of Wisconsin–Madison (2018–0103) and informed consent was obtained from all donors. CD3 T cells were isolated by negative selection following manufacturer's protocol (RosetteSep Human T cell enrichment cocktail, STEMCELL Technologies). Following isolation, T cells were plated at 1 million cells/mL in either ImmunoCult XF T cell Expansion Medium (STEMCELL Technologies) and activated with either 25µL/mL StemCell αCD2/αCD3/αCD28, 10µL/mL TransAct αCD3/αCD28, or αCD3/αCD28 Dynabeads (Gibco™) at 1:1 and 2:1 bead-to-T cell ratios following manufacturer's instructions. To focus on the specific impacts of media and activating antibodies on T cell metabolism and metabolic shift kinetics, no cytokine was supplemented to activated T cells. 200,000 T cells were sampled from each activation conditions and imaged using OMI at 24, 48, and 72 hours. αCD3/αCD28 Dynabeads were removed

from the activated T cell samples immediately prior to imaging following manufacturer's instruction to avoid bead interference with imaging.

Virus-free anti-GD2 CAR T activation time course

CD3 T cells were isolated from peripheral blood of healthy donors by negative selection as previously described. Following isolation, T cells were plated at 1 million cells/mL in either ImmunoCult XF T cell Expansion Medium (STEMCELL Technologies) supplemented with 200U/mL IL-2 (Peprotech) or TexMACS Cell Culture Medium (Miltenyi Biotec) supplemented with 10ng IL-7 (Biotechne), and stimulated with 25 μ L/mL ImmunoCult Human α CD2/ α CD3/ α CD28 T cell Activator (STEMCELL Technologies) or 10 μ L/mL T-cell TransAct α CD3/ α CD28 (Miltenyi Biotec), respectively. These two combinations of activating antibodies and culture media yielded two activation methods, Imm and Tex respectively. T cells were activated for different durations, ranging from 12 hours up to 72 hours. The metabolic profile of activated T cells was characterized with OMI while cell cycle stage and proliferative capacity were analyzed via flow cytometry of Hoechst and Ki67 staining, respectively. Cell cycle stages were gated based on median fluorescence intensity (MFI) of Hoechst stain that reflected cellular DNA content. These activated T cells were then electroporated with CRISPR/Cas9 machinery to express anti-GD2 CAR transgene as previously described⁽⁹⁾. Following electroporation, Imm- and Tex-activated anti-GD2 CAR T cells were expanded for 7 days in ImmunoCult XF T cell expansion media supplemented with 500U/mL IL-2 or TexMACS media supplemented with 10ng IL-7, respectively. After expansion, CAR T cells were harvested; percent CAR positivity was quantified via flow cytometry of 1A7 anti-14G2A antibody (National Cancer Institute, Biological Resources Branch) conjugated to APC using a Lightning Link APC Antibody Labeling kit (Novus Biologicals) to evaluate genome editing efficiency as previously described (Extended Data Fig. 2A) (9,⁴³).

CAR T cell expansion and flow cytometry characterization

T cells from 3 donors were activated with either Imm or Tex method prior to electroporation with CRISPR/Cas9 machinery to express anti-GD2 CAR as previously described (9,⁴³). Following electroporation, CAR T cells were expanded in either ImmunoCult XF or TexMACS media for 7 days in a 37C, 5% CO₂ humidified incubator. During expansion, T cell media were supplemented with 500U/mL IL-2 (Peprotech), 10ng/mL IL-7 (BioTechne), 10ng/mL IL-7 (BioTechne)+ 10ng/mL IL-15 (BioTechne) (IL-7/IL-15 low), 10ng/mL IL-7 (BioTechne) + 1ng/mL IL-15 (BioTechne) (IL-7/IL-15 low), or 10ng/mL IL-7 (BioTechne) + 100U/mL IL-2 (Peprotech) (IL-7/IL-2). Cells were counted every 2 days and adjusted to 1 million cells/mL. CAR T cell phenotypes and metabolism were analyzed with flow cytometry and OMI after 7 days of expansion. Flow cytometry was performed on an Aurora spectral cytometer (Cytek) as previously described (Extended Data Fig. 9) (9,⁴³). Briefly, T cells were stained and analyzed for expression of 16 markers (anti-GD2 CAR, TCR $\alpha\beta$, CD4, CD8, CD45RA, CD45RO, CD62L, CCR7, Human CD45, PD-1, LAG3, TIM3, CD39, TIGIT, CD27, CD5) with specific clones and concentrations as previously described⁽⁴³⁾. Antibodies used for validation of cell type and immunophenotyping include: CAR [1A7 anti-14G2A antibody (National Cancer Institute, Biological Resources Branch) conjugated to APC using a Lightning Link APC Antibody Labeling kit (Novus

Biologics Item # 705–0010)], TCR α/β (clone IP26, BV421-conjugated, Biolegend Item #306722, Lot B372838), CD4 (clone SK3, Spark Blue 550-conjugated, Biolegend Item # 344655, Lot B352122 and B357757), CD8 (clone SK1, BUV805-conjugated, BD Biosciences Item # 612890, Lot CD8 2140237 and 2251166), CD45RA (clone HI100, PE/Fire700-conjugated, Biolegend Item # 304171, Lot B324751, B370155, and B357757), CD45RO (clone UCHL1, BV480-conjugated, BD Biosciences Item # 566192, Lot 1169700 and 2146292), CD62L (clone DREG-56, Brilliant Violet 605-conjugated, Biolegend Item # 304833, Lot B320722, B347723, and B370276), CCR7 (clone G043H7, Spark NIR 685-conjugated, Biolegend Item #353257, Lot B315663), Human CD45 (clone 2D1, Spark Blue 574-conjugated, Biolegend Item #368558, Lot B347799), PD1 (clone EH12.2H7, BV785-conjugated, Biolegend Item #329929, Lot B318728 and B373725), LAG3 (clone 11C3C65, BV650-conjugated, Biolegend Item #369315, Lot B318076 and B368156), TIM3 (clone 7D3, BB515-conjugated, BD Biosciences Item #565568, Lot 2054090), CD39 (clone TU66, BUV737-conjugated, BD Biosciences Item #61285, Lot 2006647), TIGIT (clone A15153G, PE-Fire 640-conjugated, Biolegend Item #372743, Lot B361795 and B361796), CD27 (clone O323, BV570-conjugated, Biolegend Item #302825, Lot B345039 and B361196), and CD5 (clone UCHT2, BUV563-conjugated, BD Biosciences Item #741354, Lot 2215981). All antibodies were titrated to determine the optimal staining concentration as previously described (⁴³). In brief, the volumes of antibodies per sample are as following: 0.2 μ L 1A7 anti-14G2A CAR antibody, 1.25 μ L TCR α/β , 0.63 μ L CD4, 0.313 μ L CD8, 2.5 μ L CD45RA, 2.5 μ L CD45RO, 2.5 μ L CD62L, 0.63 μ L human CD45, 2.5 μ L PD-1, 4 μ L LAG3, 1.25 μ L TIM3, 0.63 μ L CD39, 2.5 μ L TIGIT, 1.25 μ L CD27, 0.313 μ L CD5, where one sample is defined as 250,000 cells for in vitro and 1 million cells for in vivo experiments. An expression histogram of each surface marker was generated based on single-cell MFIs normalized by donor-matched fluorescence-minus-one (FMO) control.

For in vivo treatment, three groups of CAR T cells were generated under distinct expansion conditions. Imm CAR T cells were activated in ImmunoCult XF media + 200 U/mL IL-2 for 48 hours before electroporation, and then expanded in ImmunoCult XF media + 500 U/mL IL-2 following electroporation. Tex CAR T cells were activated in TexMACS media containing 10 ng/mL IL-7 for 72 hours, electroporated, and subsequently expanded in TexMACS media + 10 ng/mL IL-7. Tex $\rightarrow$ Imm CAR T cells, on the other hand, were first activated in TexMACS media with 10 ng/mL IL-7 for 72 hours, electroporated, then expanded in ImmunoCult XF + 500 U/mL IL-2.

T cell expansion for OMI and metabolic assays

T cells from two healthy donors were activated with either 25 μ L/mL StemCell α CD2/ α CD3/ α CD28 in ImmunoCult XF media + 200U/mL IL-2 for 48 hours or 10 μ L/mL TransAct α CD3/ α CD28 in TexMACs media + 10ng/mL IL-7 for 72 hours. This activation duration (48 hours in ImmunoCult XF and 72 hours in TexMACs) represents the optimal activation duration for high CAR transgene incorporation as previously identified. T cells were collected post activation and washed twice with PBS to remove residual activating antibodies and spent media before transferring into different expansion media. T cells activated in ImmunoCult XF media were expanded in either ImmunoCult XF media + 500U/mL IL-2 (Imm) or TexMACs media + 10ng/mL IL-7 (Imm $\rightarrow$ Tex). T cells activated

in TexMACs media were expanded in TexMACs media + 10ng/mL IL-7 (Tex), ImmunoCult XF media + 500U/mL IL-2 (Tex→Imm), or Tex^{hi} media + 500U/mL IL-2 (Tex→Tex^{hi}). T cells were then expanded for 6 days, with cell count performed every two days before being collected for characterization with OMI and other metabolic assays.

OMI of T cells

T cells were plated on a 35mm Poly-D-Lysine glass bottom imaging dish (MatTek) at the density of 200,000 cells/75μL and allowed to settle for at least 15 minutes prior to imaging. Throughout the process of OMI, T cells were kept in a stage top incubator (37°C, 5% CO₂) to maintain their physiological conditions. OMI was performed on a custom-built multiphoton microscope (Ultima, Bruker) consisting of an inverted microscope body (Ti-E, Nikon) coupled to an ultrafast tunable laser source (Insight DS+, Spectra Physics). Images were acquired using time-correlated single-photon counting electronics (SPC 150, Becker & Hickl GmbH) using Prairie View Software (Bruker, v5.8). NAD(P)H and FAD were excited at 750 nm (2.5 mW) and 890nm (4.5mW), respectively, using a 40X water immersion 1.15 NA objective (Nikon) with 2.5x optical zoom, 4.8 μs pixel dwell time, 60s integration time, and image size of 256 × 256 pixels. NAD(P)H and FAD emission were separated from excitation light using a 720 long pass filter and collected using GaAsP photomultiplier tubes (H7422, Hamamatsu) through a 440/80 nm and 550/100nm bandpass filters, respectively. PerCP conjugated CAR antibody was excited at 980nm, while Alexa 647 CCR7 antibody was excited at 1200nm, respectively. Fluorescence emission of PerCP and Alexa 647 were collected with 690/50nm filter. Fluorescence intensity and lifetime images of NAD(P)H and FAD, together with immunofluorescence images of surface markers, were collected for each field of view (FOV), with 3–5 representative FOVs (~150–250 cells, ~120 μm × 120 μm) imaged per condition. The instrument response function was measured using second-harmonic generation signal from urea crystals excited at 890 nm, with full width at half-maximum of 260 ps.

OMI analysis

Fluorescence lifetime components were computed for each image pixel using SPCImage (v8.0, Becker and Hickl GmbH) by first thresholding the background, then the pixel-wise decay curves were fit to a biexponential model convolved with the instrument response function, using an iterative parameter optimization to obtain the lowest sum of the squared differences between model and data (Weighted Least Squares algorithm). The two-component exponential decay model is $I(t) = \alpha_1 e^{-t/\tau_1} + \alpha_2 e^{-t/\tau_2} + C$, where $I(t)$ is the fluorescence intensity at time t after the laser excitation pulse, τ_1 and τ_2 are the fluorescence lifetimes of the short and long lifetime components, respectively, α_1 and α_2 are the fractional contributions of the short and long lifetime components, respectively, and C accounts for background light. The mean fluorescence lifetime, $\tau_m = \alpha_1 \tau_1 + \alpha_2 \tau_2$, is the weighted average of the free- and protein-bound- fraction (67, 68). To enhance the fluorescence counts in each decay, a bin of 1 (comprising 9 neighbor pixels) and of a bin of 2 (comprising 25 neighbor pixels) were applied on NAD(P)H and FAD lifetime images, respectively. The pixel-wise optical redox ratio was calculated

as $\frac{NAD(P)H_{intensity}}{NAD(P)H_{intensity} + FAD_{intensity}}$. A customized CellProfiler pipeline (v3.1.5) was used for manual segmentation of every individual cell nucleus within a FOV, from which the individual cell border was propagated to create a whole-cell mask. A single-cell cytoplasm mask was generated by subtracting the nuclei mask from the whole cell mask. The cytoplasm mask was applied to the corresponding OMI image to compute mean values of OMI parameters for each cell cytoplasm. 13 OMI parameters were collected and quantified (Table 1). Cytoplasm size was also computed for individual cells based on cytoplasm masks.

Classification and Uniform Manifold Approximation and Projection (UMAP) based on OMI parameters were performed using custom Python scripts (Python v3.9.18). Z-score-based heatmaps were generated using custom R scripts executed in RStudio (R v3.5.3).

Extracellular flux analysis

Extracellular flux assay was performed using Seahorse XF Cell Mito Stress Test Kit (Agilent). 24 hours prior to the assay, a Seahorse cell culture 96 well-plate (Agilent) was coated with 50µL/well of 50 µg/mL Gibco Poly-D-lysine (ThermoFisher Scientific) for 1 hour, then washed with distilled water before storing overnight at 4°C. The cell culture plate was equilibrated to room temperature before cell plating. 400,000 T cells/well were plated onto Poly-D-lysine coated cell culture plate in RPMI XF media (Agilent) supplemented with 10mM glucose (Agilent) and 2mM glutamine (Agilent) following the manufacturer's protocol for seeding suspension cells in Seahorse XFp cell culture miniplate. Briefly, the T cell culture plate was centrifuged at 200g for 1 minute (no brake) and checked under the microscope to ensure even adhesion of T cells. T cells were then kept in a non-CO₂ incubator for at least 1 hour before running the assay. 1.5µM FCCP, 2.5µM Oligomycin, and 0.5µM Antimycin A/Rotenone were loaded into port A, B, and C respectively as metabolic inhibitors. Oxygen consumption rate (OCR) and extracellular acidification rate (ECAR) were measured with an XF96 Extracellular Flux Analyzer (Seahorse Bioscience).

Lactate measurement

At harvest, 1 million CAR T cells for each condition were plated in 1mL fresh media (similar to their expansion condition) in a tissue culture treated 24-well plate (VWR). After 24 hours, spent media was collected for each condition and lactate secretion was assayed in duplicate or triplicate using either Lactate Colorimetric/Fluorometric kit (abcam) or Lactate-Glo (Promega) following manufacturers' protocols. Lactate in base TexMACS and ImmunoCult media were measured as the baseline controls. For abcam's Lactate Colorimetric/Fluorometric kit (abcam), lactate secretion for each condition was measured based on the fluorescence intensity at 535/587nm excitation/emission using Tecan M1000 plate reader. Fitting and extrapolating of standard curve and sample measurements were performed in GraphPad Prism v10 (linear curve fitting).

ATP production and cellular reducing capacity assays

Pan T cells were isolated (Miltenyi) from PBMCs (Leukopak, StemCell Technologies) and plated in TexMACS or ImmunoCult at 7×10^5 cells/ml in a 6-well plate. Cells were then subsequently activated with TransAct (Miltenyi), including a non-activated control. After

activation, samples were taken and incubated with RealTime-GloTM MT (Promega) per manufacturer's instructions in a CO₂ and temperature-controlled plate reader (Tecan Spark Cyto). Luminescence was recorded every hour for 74 hours to quantify cellular reducing capacity. Additionally, daily samples were obtained, and ATP levels were determined using CellTiter-Glo[®] from Promega, following the manufacturer's instructions.

ATP production, metabolic cofactor, and dehydrogenase assays

T cells expanded in Imm, Tex, Tex→Imm, Tex→Texhi conditions were collected after 6 days of expansion (day 9 of mock manufacturing process) for OMI imaging and parallel characterization of ATP production as described above. Metabolic cofactors and their ratios (NAD, NADH, NADP, NADPH) were measured using NAD/NADH-GloTM and NADP/NADPH-GloTM assays (Promega) following the manufacturer's instructions. Dehydrogenase-GloTM Detection System (Promega) was used to determine activity levels of several dehydrogenases (LDH, GAPDH, IDH, 6DGPH, G6PDH) in these T cell samples following the manufacturer's instructions.

Mouse CHLA-20 xenograft and CAR T cell treatment

All animal experiments were approved by the University of Wisconsin-Madison Animal Care and Use Committee (ACUC protocol M005915). All mice were housed with 12-hour dark/light cycle at 22–23°C and 40–50% humidity. Establishment of CHLA-20 xenograft and subsequent CAR T cell treatments were performed as previously described^(9,43). Briefly, male and female NOD-SCID- γ c^{-/-} (NSGTM) mice (9–25 weeks old; Jackson Laboratory) received 10 million AkaLUC-GFP CHLA-20 GD2⁺ human neuroblastoma cells via subcutaneous flank injection to establish tumors. Tumor flux was measured with IVIS imaging one day prior to CAR T cell treatment, and mice were divided into three treatment groups to control for equally distributed tumor flux in each group. 4 million anti-GD2 CAR⁺ cells were injected into the tail vein of tumor-bearing each mouse. Tumor flux was monitored with IVIS imaging every 3–4 days. During CAR T treatment duration, mice also received 100,000 IU of human IL-2 (National Cancer Institute, Biological Resources Branch) subcutaneously on day 0 and post imaging. Total flux was calculated using Living Image Software (PerkinElmer, v4.7.4) as radiance (photons/second) in each pixel integrated over tumor area (cm²) $\times 4\pi$. To normalize for background signal, the minimum flux value was subtracted from each image.

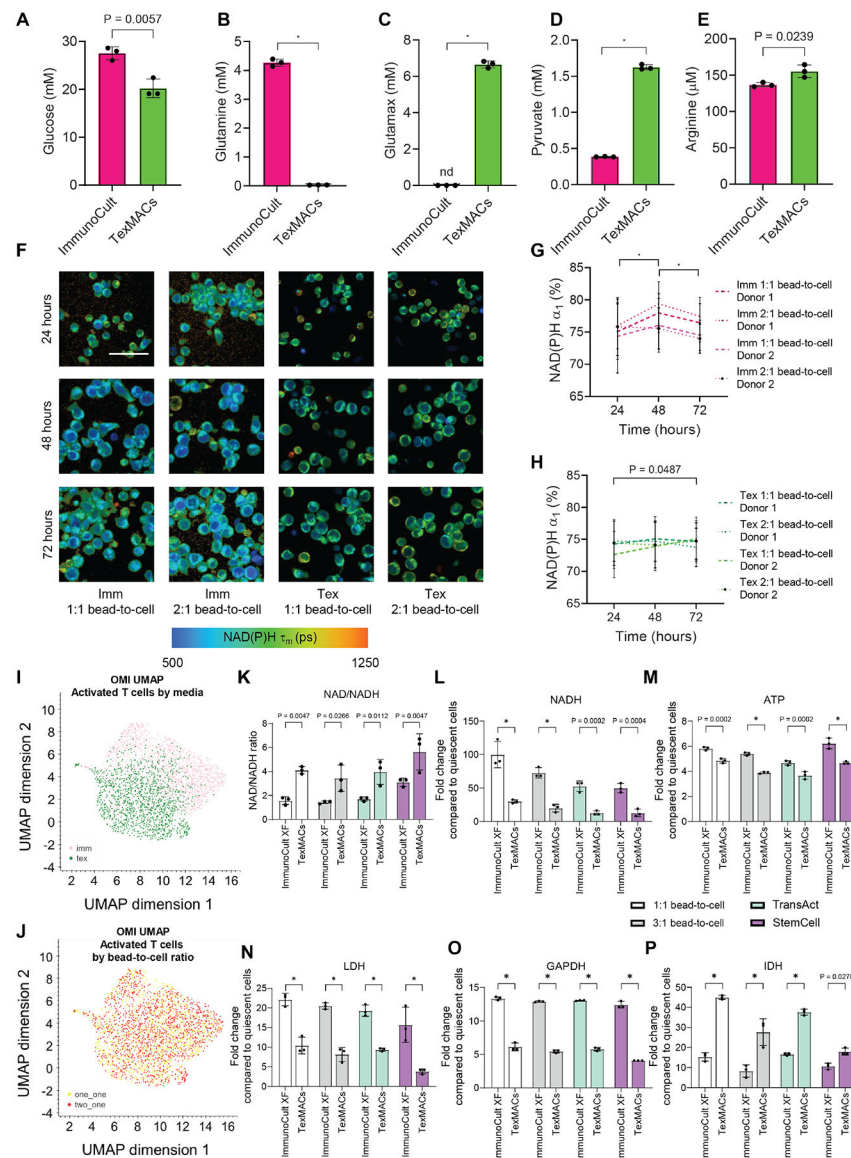
In vitro cytotoxicity assay

To measure antitumor cytotoxicity of anti-GD2 CAR T cells, GFP⁺M21 melanoma or MG63 at a density of 5,000 cells per well in 200 μ L of DMEM supplemented with 10% FBS and 1% penicillin-streptomycin. After 24 hours of incubation, Tex→Imm and Imm CAR T cells were added to the coculture at CAR⁺ T-cell-to-cancer-cell ratios of 1:1, 1:2, 1:5, and 1:10. The plate was centrifuged at 100g for 1 minute to ensure cell contact and then transferred to the IncuCyte S3 live-cell imaging system (Sartorius). Time-lapse images were captured every 60 minutes for a duration of 72 hours. The images were analyzed using IncuCyte S3 Cell-by-Cell software, with tumor cell viability assessed by the reduction in green fluorescent signal or in the percentage of GFP⁺ cells within the field of view over time.

Statistical analysis

All statistical analysis was performed in Prism GraphPad v10, with appropriate statistical tests chosen based on data features. $p < 0.05$ was chosen as the threshold for statistical significance. Multiple comparisons were adjusted with post-hoc tests. Glass's delta for effect size was calculated as $\frac{|\mu_{treatment} - \mu_{control}|}{s_{control}}$ where $\mu_{treatment}$ and $\mu_{control}$ representing the means of treated and control groups, respectively; and $\sigma_{control}$ being the standard deviation of the control group. MFI of surface markers) were normalized to donor-match control groups to account for any variations in laser power. Details on the number of sampled units and specific statistical tests for each experiment were reported in corresponding figure legends.

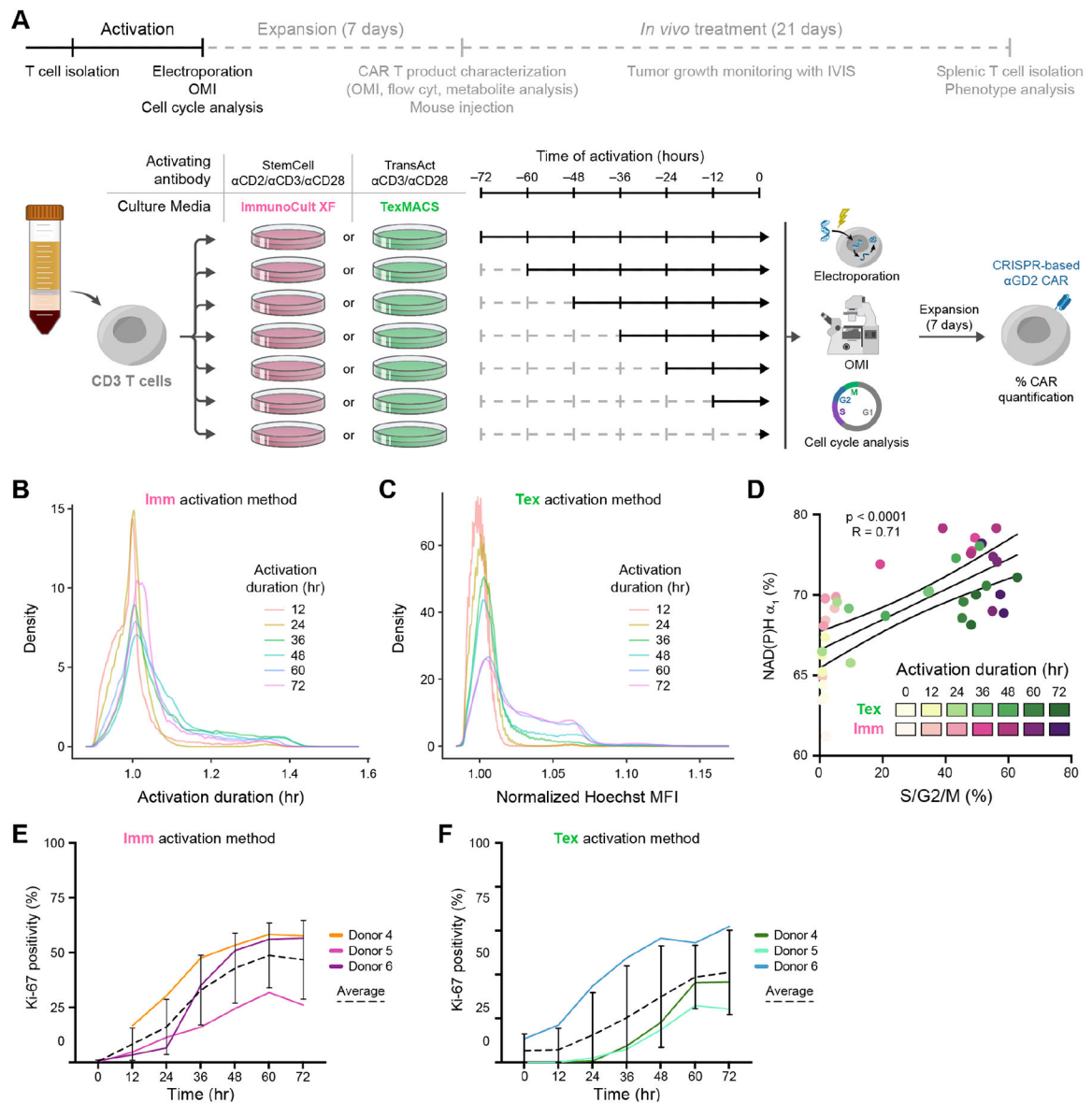
Extended Data



Extended Data Fig. 1: OMI and cofactor/enzyme activity assessment revealed the impact of culture media on activation kinetics and metabolism of T cells stimulated with α CD3/ α CD28 Dynabeads.

OMI and cofactor/enzyme activity assessment revealed the impact of culture media on activation kinetics and metabolism of T cells stimulated with α CD3/ α CD28 Dynabeads. (A-E) Liquid chromatography-mass spectrometry analysis revealed significantly different composition in ImmunoCult XF and TexMACs media. Concentration of (A) glucose, (B) glutamine, (C) glutamax, (D) pyruvate, and (E) arginine quantified in ImmunoCult XF and TexMACs media. $n = 3$ replicates, two-sided unpaired t test. (F) Representative NAD(P)H τ_m images from T cells activated with α CD3/ α CD28 Dynabeads at 1:1 and 2:1 bead-to-T cell ratio in ImmunoCult XF and TexMACs media. α CD3/ α CD28 Dynabeads were removed immediately prior to imaging. (G-H) T cells activated in ImmunoCult XF and TexMACs

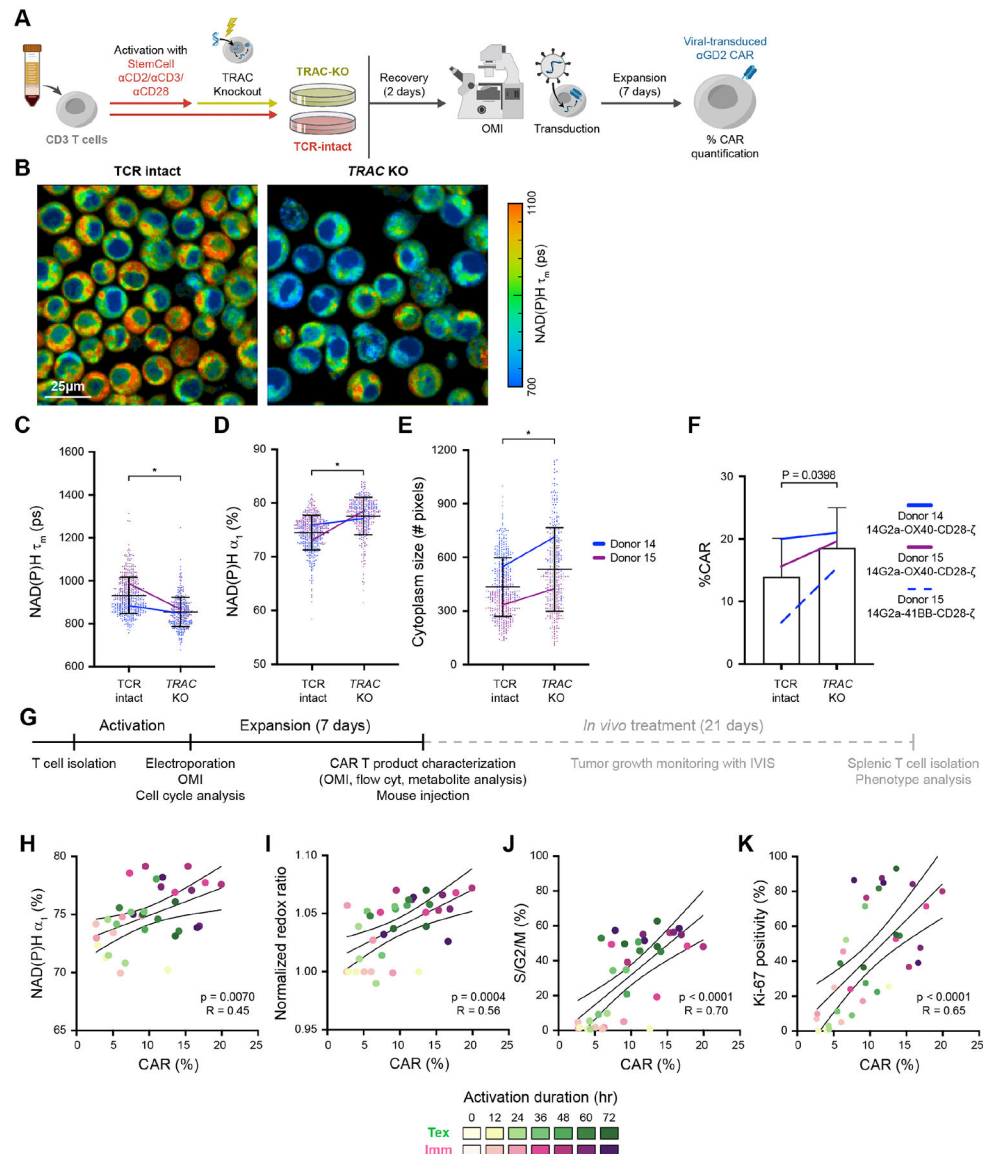
media demonstrated different activation kinetics. NAD(P)H α_1 quantification of T cells from two healthy donors activated with α CD3/ α CD28 Dynabeads in (G) ImmunoCult XF and (H) TexMACs media at 24-, 48-, and 72-hours post activation. For (G) $n = 449, 437$, and 432 activated T cells from 3 donors in ImmunoCults media at 24, 48, and 72 hours. For (H), $n = 475, 446$, and 410 activated T cells from 3 donors in TexMACs media at 24, 48, and 72 hours. For (G-H) two-way ANOVA with Tukey's posthoc test for multiple comparisons. (I-J) UMAP based on Euclidean distances of 11 OMI parameters (NAD(P)H τ_m , τ_1 , τ_2 , α_1 , intensity, FAD τ_m , τ_1 , τ_2 , α_1 , intensity, redox ratio) and cell size of T cells from two healthy donors activated with α CD3/ α CD28 Dynabeads, color coded based on (I) culture media and (J) bead-to-T cell ratio. $n = 2649$ cells. (K) NAD/NADH ratio, and fold increase in (L) NADH concentration, (M) ATP production, (N) LDH, (O) GAPDH, and (P) IDH activity measured from T cells activated with α CD3/ α CD28 Dynabeads (at 1:1 and 3:1 bead-to-T cell ratio), StemCell, or TransAct antibodies for 72 hours in ImmunoCult XF or TexMACs media. The fold increase was calculated as compared to quiescent cells. $n = 3$ replicates/condition, two-way ANOVA with Sidak's posthoc test for multiple comparisons. Scale bar is 50 μ m. Bars are mean $\pm$ SD. * $p < 0.0001$.



Extended Data Fig. 2:

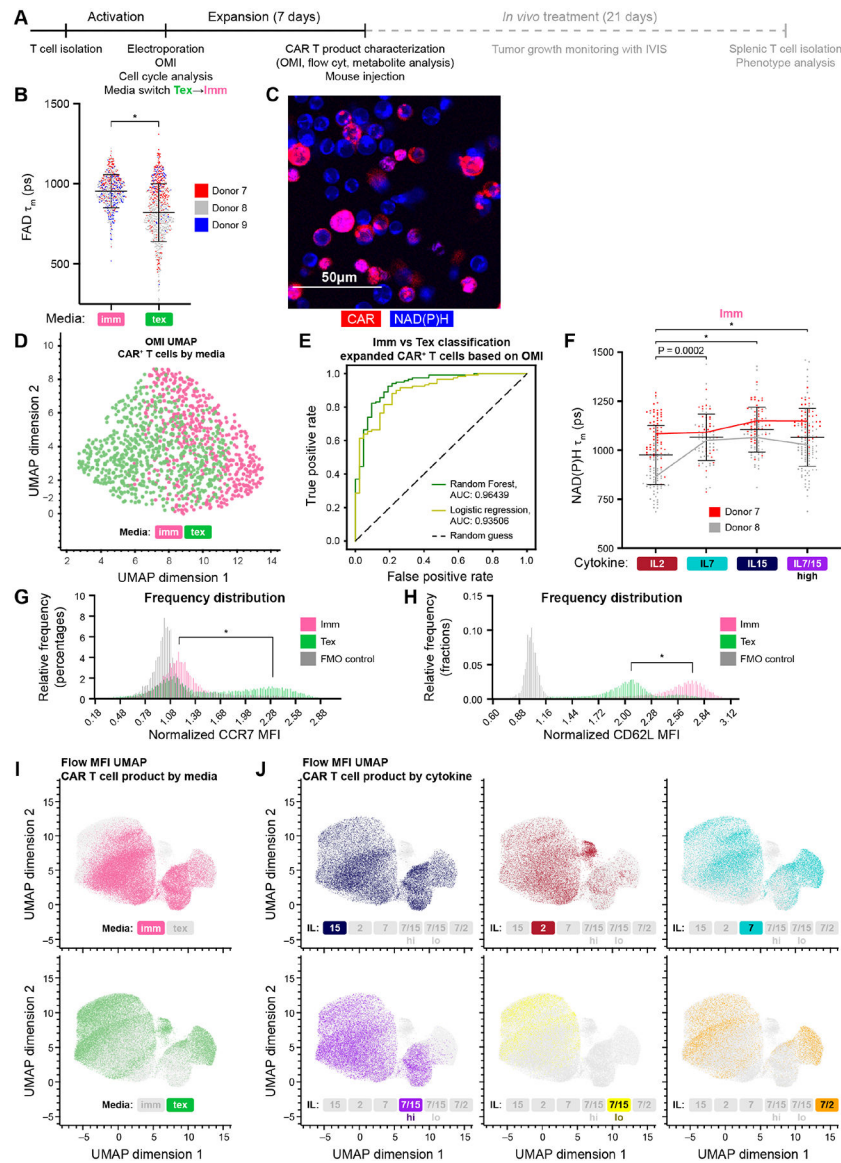
T cells progressed through cell cycle and proliferated upon activation. T cells progressed through cell cycle and proliferated upon activation. (A) Experimental design and timeline. CD3 T cells were isolated from 3 healthy donors and activated with StemCell α CD2/ α CD3/ α CD28 in ImmunoCult XF media (Imm) or TransAct α CD3/ α CD28 in TexMACS media (Tex). Cells were divided into 7 groups with duration of activation ranging from 0 hours (quiescent T cells) to 72 hours prior to electroporation to introduce the anti-GD2 CAR transgene. When activated T cells were collected for electroporation, their metabolic features were characterized using OMI. Meanwhile, their cell cycle stage and proliferation capacity were assessed using flow cytometry of Hoechst 33342 and Ki-67 staining. These activated T cells were then electroporated to incorporate the anti-GD2 CAR transgene and expanded in corresponding culture media. Genome editing efficiency was determined after 7 days of expansion with GD2 CAR antibody using flow cytometry. (B,

C) T cells progressed through cell cycle upon activation. Density plots of Hoechst MFI in T cells activated with (B) Imm and (C) Tex methods. For each donor, Hoechst MFI was normalized to the average of 12-hour-activated group. $n = 393,999$ cells and $370,802$ cells for Imm and Tex activation methods, respectively. (D) Correlation between NAD(P)H α_1 and cell cycle stage (% cells in S/G2/M phase) at electroporation. Each dot represents one sample average, color coded based on the method of activation (Imm or Tex) and duration of activation (0–72 hours). $n = 42$ samples, Pearson R analysis. (E, F) T cell proliferation (% Ki-67⁺) following activation with (E) Imm method or (F) Tex method. $n = 3$ donors, Brown-Forsythe and Welch ANOVA test with Dunnett's post hoc test for multiple comparisons. Bars are mean $\pm$ SD. Color lines connected donor-match averages.



Extended Data Fig. 3: T cell characteristics at the time of CAR gene transfer correlated with transgene incorporation efficiency.

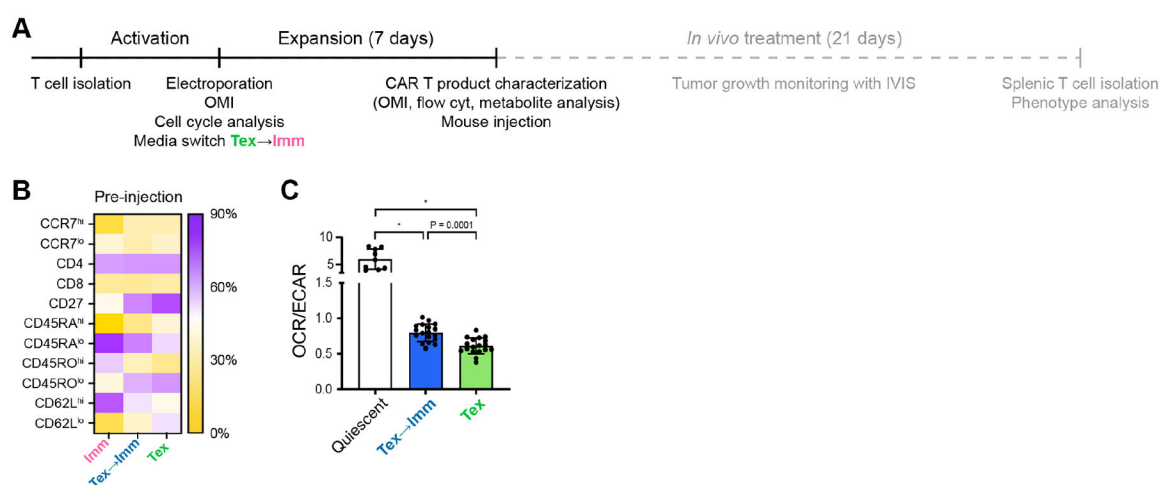
T cell characteristics at the time of CAR gene transfer correlated with transgene incorporation efficiency. (A) Experimental setup for viral transduction experiment. T cells were isolated from peripheral blood of two healthy donors and activated with Imm method. Activated CD3 T cells underwent electroporation with Cas9 ribonucleoproteins targeting the human *TRAC* locus to knockout the T cell receptor (*TRAC* KO) 2 days prior to transduction with retrovirus to express anti-GD2 CAR receptor using two constructs (OX40-CD28-CAR or 41BB-CAR) as previously described⁹. OMI was performed immediately before transduction. Transduction efficiency was quantified as %CAR⁺ after 7 days of expansion. (B) Representative NAD(P)H τ_m images of TCR-intact and *TRAC* KO T cells at transduction. OMI parameters including (C) NAD(P)H τ_m , (D) NAD(P)H α_1 , and (E) cytoplasm size of cells with intact T cell receptor (TCR-intact) and *TRAC* KO T cells at transduction. $n = 480$ TCR intact and 374 *TRAC* KO cells from 2 independent donors, two-sided Mann-Whitney test. (F) Transduction efficiency (% CAR) of TCR-intact and *TRAC* KO cells at the end of the manufacturing process. $n = 17$ samples from 2 donors and 2 anti-GD2 CAR constructs, unpaired two-sided T test. (G) Experimental timeline of electroporation experiment. (H–K) Correlation between cell characteristics (H) NAD(P)H α_1 , (I) normalized redox ratio, (J) %S/G2/M and (K) %Ki-67⁺ at electroporation and genome editing outcome (% CAR). Each dot represents one sample average, color coded based on the method of activation (Imm or Tex) and duration of activation (0–72 hours). $n = 36$ samples, Pearson R analysis. Quiescent T cells with 0-hour activation duration did not undergo genome editing. Bars are mean $\pm$ SD. Color lines connected donor-match averages. * $p < 0.0001$.



Extended Data Fig. 4: CAR T cells expanded in Imm and Tex media displayed distinct metabolic and phenotypic features.

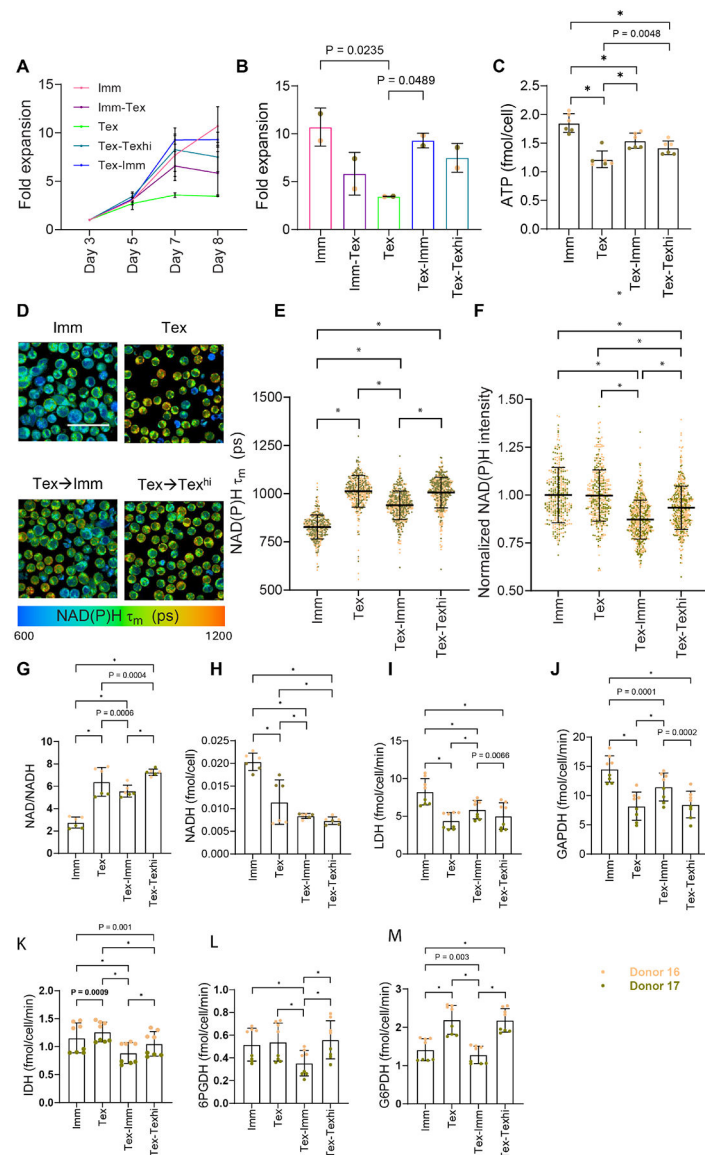
CAR T cells expanded in Imm and Tex media displayed distinct metabolic and phenotypic features. (A) Experimental timeline. (B) Quantification of FAD mean lifetime ($FAD \tau_m$) of CAR T cells expanded in either ImmunoCult XF or TexMACs media. $n = 666$ and 751 CAR T cells from 3 independent donors expanded in ImmunoCult XF and TexMACs media supplemented with various cytokine combinations, two-sided Mann-Whitney test. (C–E) OMI metabolic profiles of CAR⁺ T cells differed based on expansion media. (C) Representative image of CAR⁺ T cells identified based on PerCP conjugated GD-2 CAR antibody (red). (D) UMAP based on 13 OMI parameters of CAR⁺ T cells expanded in Imm or Tex media. (E) ROC curves and AUCs of three models based on OMI parameters (Table 1) to classify CAR⁺ T cells by expansion media (Imm vs. Tex); $n = 494$ cells for training (80%), $n = 123$ cells for testing (20%). (F) Quantification of NAD(P)H τ_m from CAR T cells expanded in ImmunoCult XF media supplemented with 500U/mL IL-2

compared to other cytokine combinations, $n=123$, 83, 96, and 129 CAR T cells expanded in IL-2, IL-7, IL-17, and IL-7+IL-15(high) supplemented media from 2 donors, ANOVA with Kruskal-Wallis test. MFIs of (G) CCR7 and (H) CD62L of CAR⁺ T cells expanded in either Imm or Tex media supplemented with different cytokine cocktails, normalized by fluorescence-minus-one (FMO) controls. ($n = 87,317$ cells from 3 independent donors). UMAP based on Euclidean distances of MFIs of 6 surface markers (CD27, CD45RO, CD62L, CD28, CD45RA and CCR7) of CAR⁺ T cells expanded in Imm and Tex media, color coded based on (I) culture media or (J) supplemented cytokines. Dots are CAR⁺ T cells. $n = 87,317$ cells from 3 independent donors. Bars are mean $\pm$ SD. * $p < 0.0001$.



Extended Data Fig. 5: CAR T cell phenotypes and metabolic profile before in vivo treatment course.

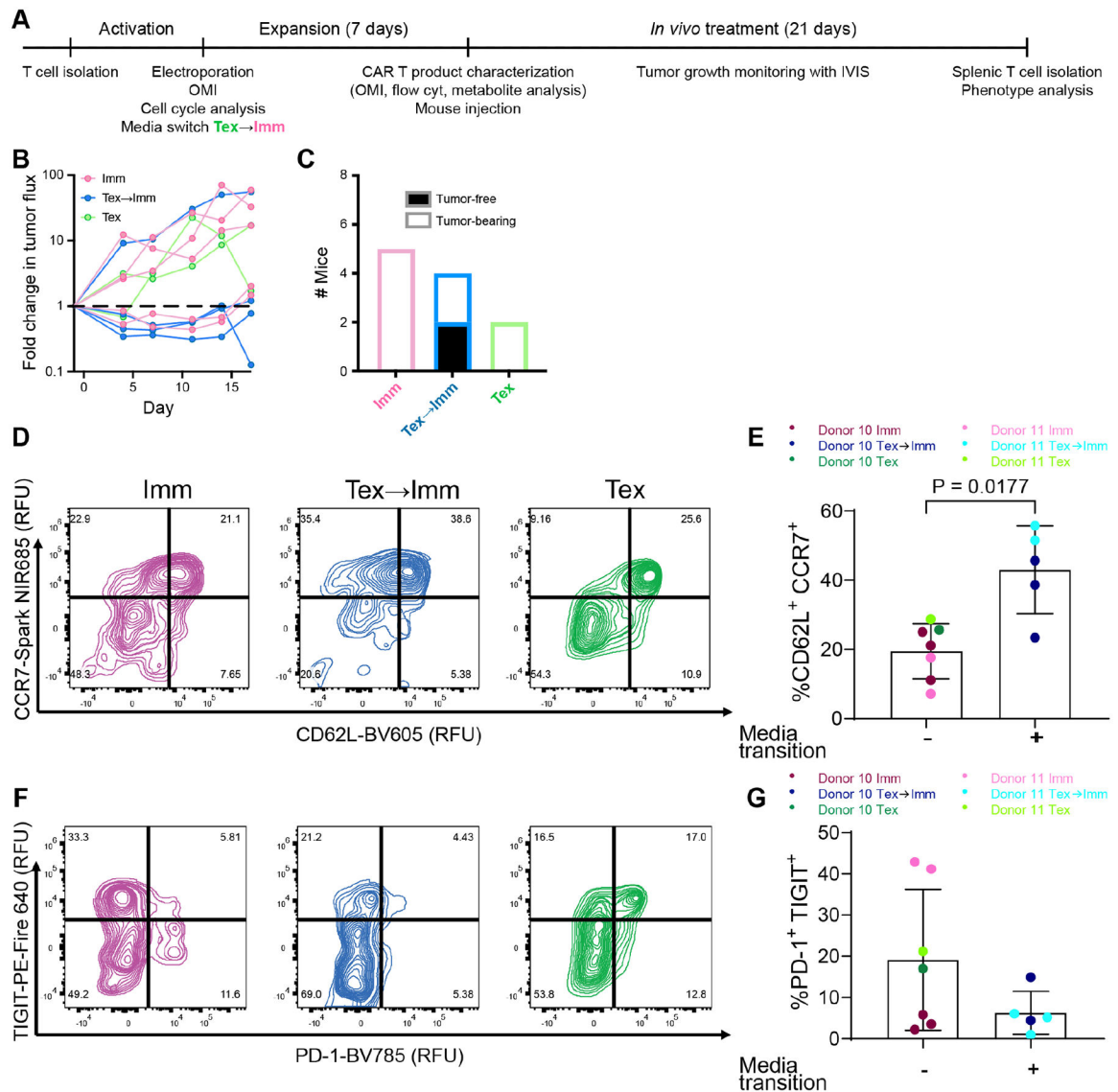
CAR T cell phenotypes and metabolic profile before in vivo treatment course. (A) Experimental timeline. (B) Phenotypes of CAR⁺ T cells expanded in Imm, Tex→Imm, and Tex conditions based on expression of 7 surface markers. (C) Extracellular flux analysis of baseline oxygen consumption rate (OCR) and extracellular acidification rate (ECAR) from CD3 T cells that were unactivated (quiescent) or activated and expanded in Imm or Tex→Imm condition. $n = 9-18$ samples/condition. Brown-Forsythe and Welch ANOVA tests with Dunnett's T3 post hoc test for multiple comparisons. Bars are mean $\pm$ SD. * $p < 0.0001$.



Extended Data Fig. 6: Expansion conditions shaped T cell expansion capacity and metabolism. (A-B) Imm and Tex-Imm T cells exhibited high expansion capacity.

Expansion conditions shaped T cell expansion capacity and metabolism. (A-B) Imm and Tex-Imm T cells exhibited high expansion capacity. Fold expansion of T cells from two healthy donors under various expansion conditions (A) throughout and (B) on day 8 of the expansion process $n = 4-6$ replicates/condition across two donors, two way ANOVA test with Tukey's posthoc test for multiple comparisons. (C) ATP production by T cells under several expansion conditions. (D-M) Media transition from TexMACs to ImmunoCult XF (Tex-Imm) shifted T cell metabolism towards balanced oxidative phosphorylation while maintaining ATP production. (D) Representative NAD(P)H τ_m images of T cells on day 9. (E) Quantification of NAD(P)H τ_m and (F) normalized NAD(P)H intensity of expanded T cells from two healthy donors on day 9. $n = 361-580$ cells/conditions across 2 donors, color coded by donors. (G-H) Metabolic cofactors, (I-M) dehydrogenases activity measured from expanded T cells of two healthy donors on day 9. $n = 6$ replicates across 2 donors.

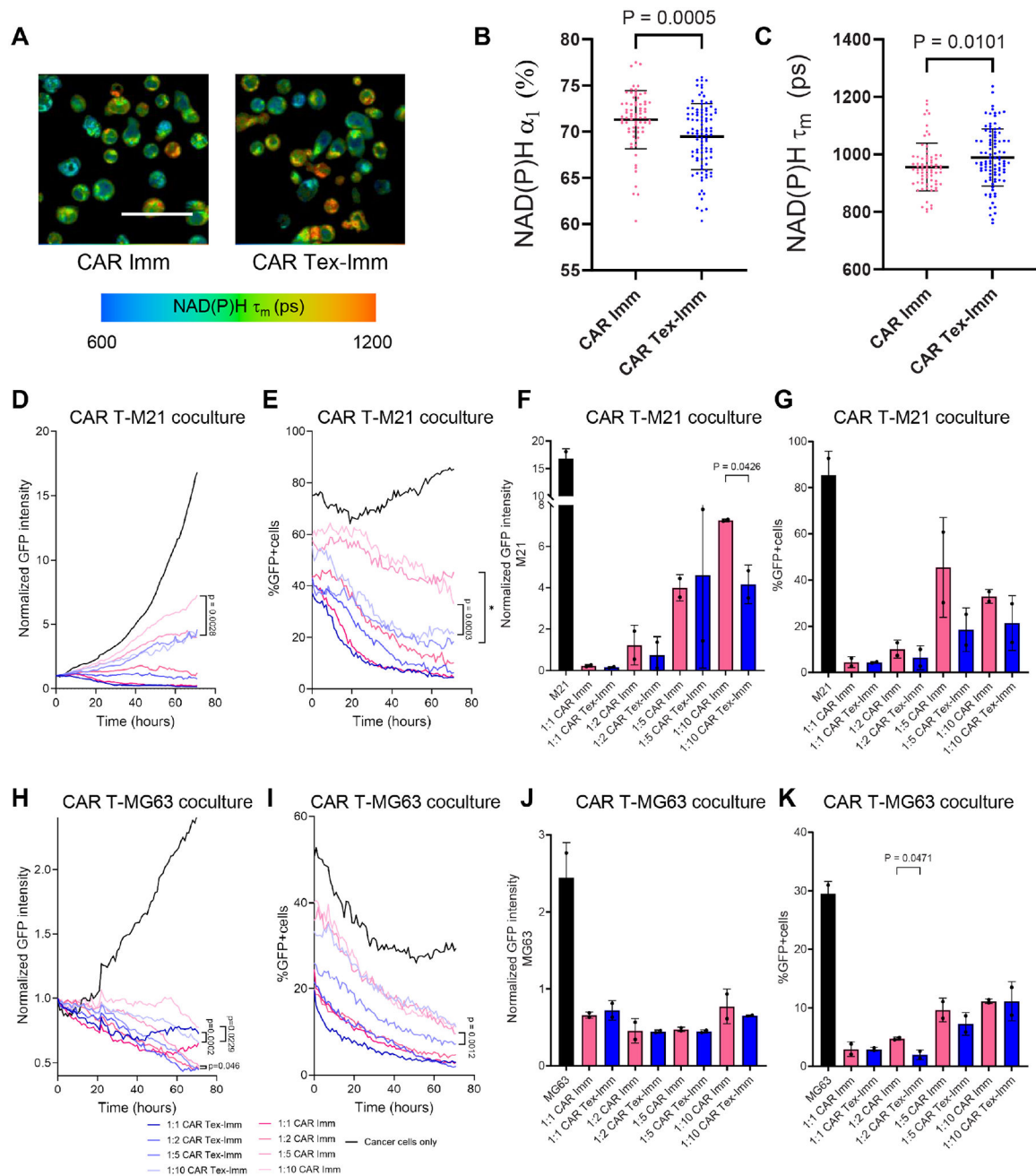
Brown-Forsythe and Welch ANOVA test with Dunnett's T3 post hoc test for multiple comparisons. Bars are mean $\pm$ SD. * $p < 0.0001$.



Extended Data Fig. 7: In vivo treatment response and phenotyping analysis of CAR T cells post in vivo treatment.

In vivo treatment response and phenotyping analysis of CAR T cells post in vivo treatment. (A) Experimental timeline. (B) Fold change in tumor flux (measured with IVIS imaging) in NSG tumor-bearing mice after CAR T cell treatment. Dash line represents no change in tumor flux (fold change = 1). (C) CAR T treatment outcomes on day 17 post treatment. (D-G) Expression of stem central memory (D-E) and exhaustion phenotypes (F-G) in CAR T cells isolated from treated NSG mouse spleen at day 21 post-treatment. (D, F) Representative flow histograms from one individual mouse in each treatment group. (E, G) Quantification of stem central memory (CD62⁺ CCR7⁺) and exhausted (PD-1⁺ TIGIT⁺)

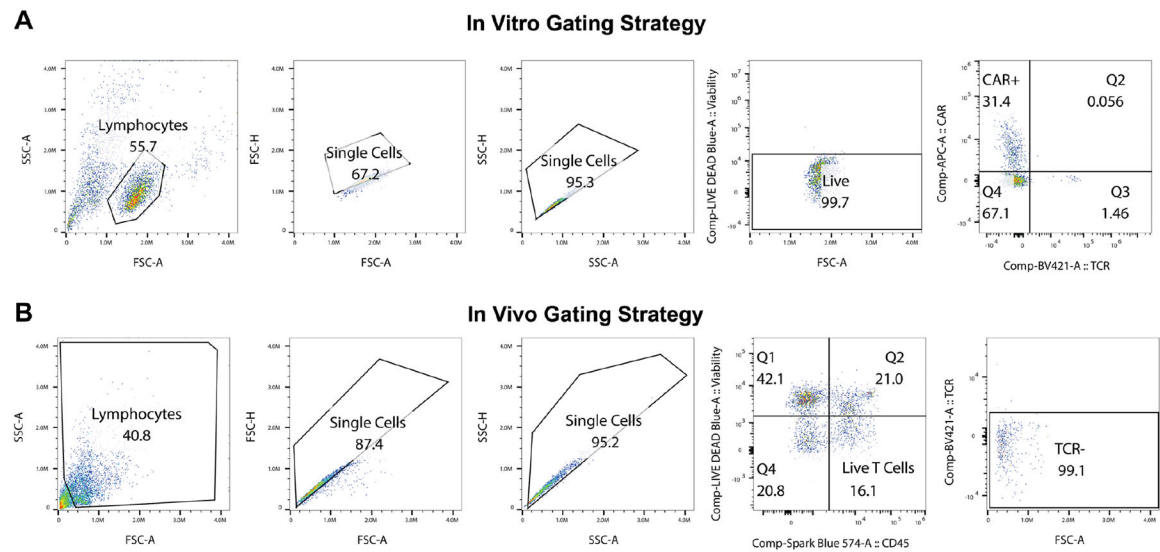
populations in treated mouse spleen. $n = 12$ mice, color coded by CAR T treatment groups. Two-sided non-parametric Mann-Whitney test. Bars are mean $\pm$ SD.



Extended Data Fig. 8: Tex→Imm CAR T cells demonstrated higher in vitro cytotoxicity than Imm CAR T cells against two cancer models at low effector-to-target ratios.

Tex→Imm CAR T cells demonstrated higher in vitro cytotoxicity than Imm CAR T cells against two cancer models at low effector-to-target ratios. (A) Representative NAD(P)H τ_m images of donor-matched CAR T cells expanded in Imm and Tex-Imm conditions. (B-C) Quantification of NAD(P)H α_1 and NAD(P)H τ_m from Imm and Tex-Imm CAR T cells at the end of manufacturing. (D-G) GFP+ GD2+ M21 melanoma and (H-K) GFP+

GD2+ MG63 osteosarcoma were cocultured with donor-matched GD2-CAR Imm and CAR Tex-Imm T cells at various effector-to-target ratios (1:1, 1:2, 1:5, and 1:10) for 72 hours. (D-E) Normalized GFP intensity and percentage of GFP+ cells in the coculture of CAR T cells and M21 melanoma throughout 72 hours and (F-G) at 72-hour timepoint. (H-I) Normalized GFP intensity and percentage of GFP+ cells in the coculture of CAR T cells and MG63 osteosarcoma throughout 72 hours and (J-K) at 72-hour timepoint. For (D-E) and (H-I) n = 2 well/condition x 72 time points, Friedman test (one-way repeated measures analysis of variance by ranks) with Dunn’s posthoc test for multiple comparisons. For (F-G) and (J-K), n = 2 replicates/condition, unpaired t-test. Bars are mean ± SD. * p < 0.0001. Scale bar is 50µm.



Extended Data Fig. 9: Gating strategy for analysis of spectral immunophenotyping flow cytometry of CAR T cells post-manufacturing (A) or isolated from treated NSG mouse spleen post-treatment (B).

Extended Data Table. 1: Glass’s deltas (Δ) showing maximal effect sizes of Imm and Tex activation methods on T cell metabolism ($\text{NAD(P)H } \tau_m$ and $\text{NAD(P)H } \alpha_1$). Glass’s deltas (Δ) showing maximal effect sizes of Imm and Tex activation methods on T cell metabolism ($\text{NAD(P)H } \tau_m$ and $\text{NAD(P)H } \alpha_1$). For each donor and activation method, Glass’s deltas were calculated based on $\text{NAD(P)H } \tau_m$ and $\text{NAD(P)H } \alpha_1$ at the time point (12, 24, 36, 48, 60, or 72 hours of activation duration) of maximal differences compared to donor-matched quiescent T cells (0-hr activation duration) in respective media (Imm or Tex).

Variable	$\Delta\text{NAD(P)H } \tau_m$		$\Delta\text{NAD(P)H } \alpha_1$	
Activation method	Imm	Tex	Imm	Tex
Donor 1	2.61	1.56	3.76	2.50
Donor 2	3.55	1.66	5.53	3.35
Donor 3	2.18	1.57	4.64	2.47

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Data availability:

All data supporting the findings of this study have been deposited and are available on Zenodo (⁶⁸).

Code availability:

All relevant analysis codes have been deposited on Zenodo (⁶⁸).

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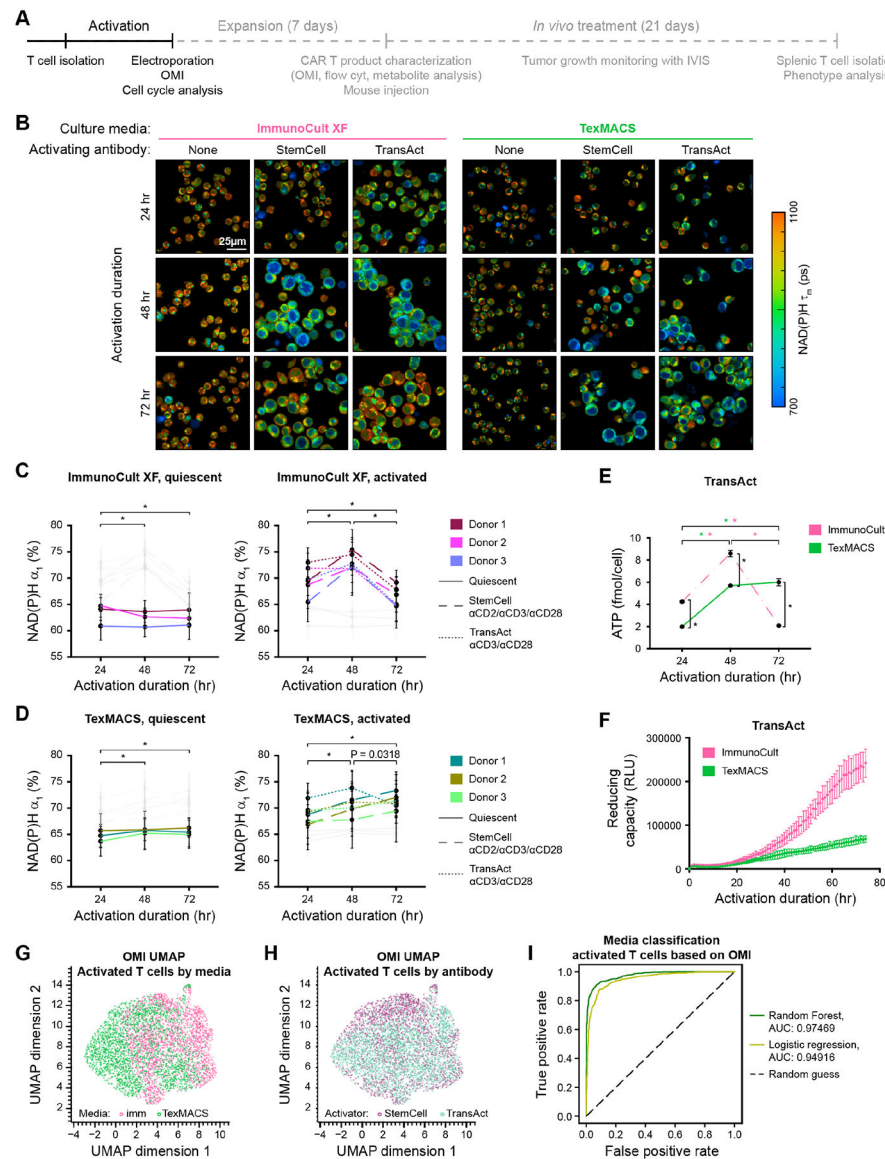


Fig. 1: Activating media, not activating antibody, determines CD3 T cell metabolism. (A) Experimental timeline. (B) Representative NAD(P)H mean lifetime (NAD(P)H τ_m) images. (C, D) quantification of NAD(P)H τ_m of quiescent (left) and activated (right) CD3 T cells in either (C) ImmunoCult XF or (D) TexMACS media. For (C), left panel: $n = 511$, 593, and 571 biologically independent quiescent T cells at 24, 48, and 72 hours; right panel: $n = 1467$, 1320, and 1588 T cells activated with either StemCell or TransAct antibody at 24, 48, and 72 hours. For (D), left panel: $n = 474$, 556, and 503 quiescent T cells at 24, 48, and 72 hours; right panel: $n = 892$, 885, and 814 activated T cells at 24, 48, and 72 hours. For (C) and (D), two-way ANOVA and adjusted multiple comparison (E) ATP production and (F) reducing potential of T cells activated with TransAct α CD3/ α CD28 antibody in ImmunoCult XF and TexMACS media. RLU: relative light unit. $n = 3$ replicates/condition/ timepoint, two-way ANOVA with Fisher's LSD post-hoc test for multiple comparison. (G, H) UMAP based on Euclidean distance projection of 14 OMI parameters (Table 1)

from T cells activated for 24–72 hours, colors representing **(G)** culture media and **(H)** activating antibody. $n = 4,923$ cells from 3 independent donors. **(I)** Receiver operating characteristic (ROC) curves and areas under the curve (AUCs) Random Forest (RF) and Logistic Regression (LR) algorithms to classify T cells activated in ImmunoCult XF versus TexMACS media (including both antibody conditions and all activation durations) based on OMI parameters. Data were randomly split into 70% for training ($n = 3,446$ cells) and 30% for testing ($n = 1,477$ cells). Bars are mean $\pm$ SD. * $p < 0.0001$.

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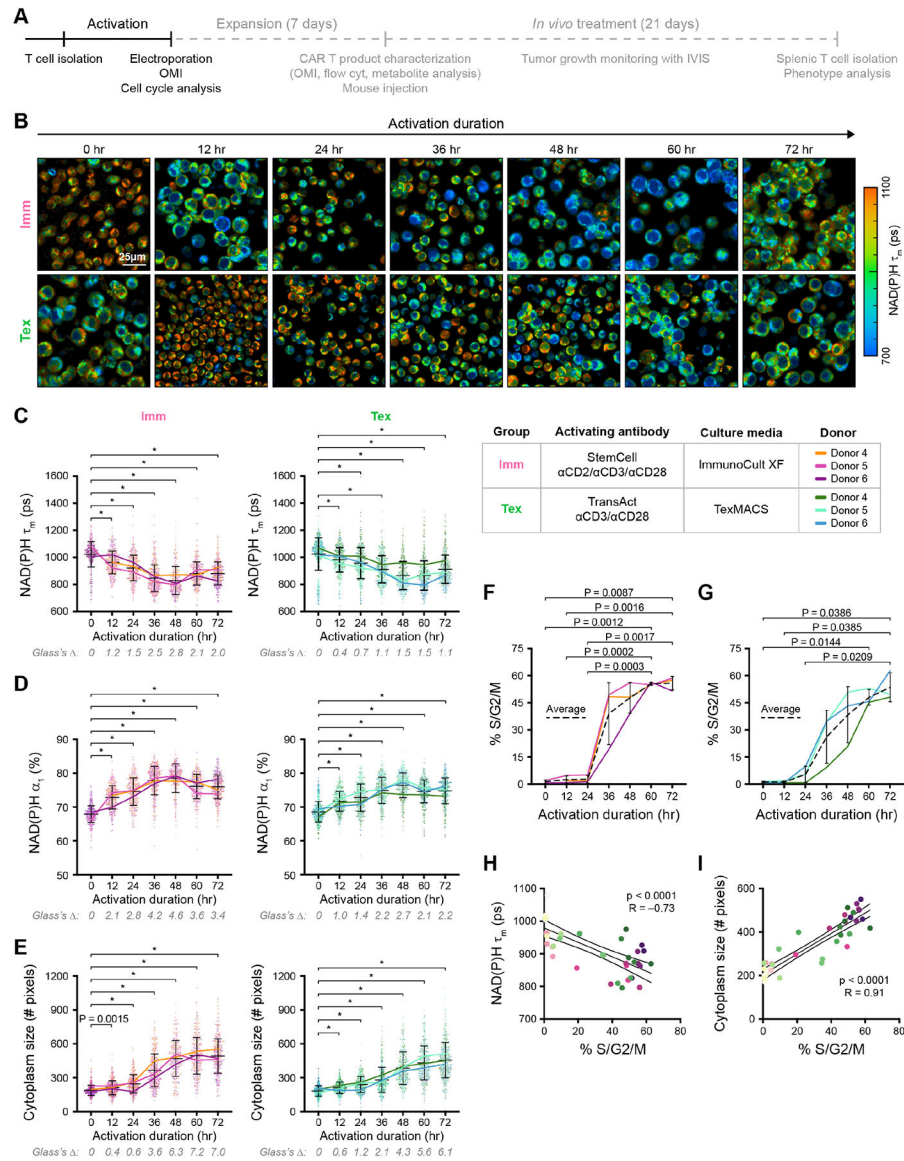


Fig. 2: OMI is sensitive to metabolic changes as T cells progress through cell cycle following activation.

(A) Experimental timeline. **(B)** Representative NAD(P)H τ_m images of T cells activated for 12–72 hours with Imm (top row) or Tex (bottom row) activation methods. Quantification of **(C)** NAD(P)H τ_m , **(D)** NAD(P)H α_1 , and **(E)** cytoplasm size of T cells activated with Imm method (left column, pink) or Tex method (right column, green). For **(C–E)**, left panel: $n = 390, 386, 354, 420, 314, 346, 391$ T cells activated for 0, 12, 24, 36, 48, 60, and 72 hours in Imm condition across 3 donors; right panel: $n = 686, 535, 469, 480, 412, 410, 450$ T cells activated for 0, 12, 24, 36, 48, 60, and 72 hours in Tex condition across 3 donors; Brown-Forsythe and Welch ANOVA test with Dunnett T3 post hoc test. Glass's deltas were calculated with respect to quiescent cells as the effect sizes of Imm and Tex activation methods on OMI parameters. **(F, G)** Percentage of T cells in S/G2/M phase following activation with **(F)** Imm or **(G)** Tex method. $n = 3$ donors, Brown-Forsythe and Welch ANOVA test with Dunnett T3 post hoc test. **(H, I)** Correlation between OMI parameters,

including **(H)** NAD(P)H τ_m and **(I)** cytoplasm size, and the percentage of cells in S/G2/M phase. $n = 42$ samples from 3 donors, Pearson R analysis. **(B–G)** Color solid lines represent donor averages. **(H, I)** Each dot represents one sample average, color coded based on the method (Imm or Tex) and duration of activation (0–72 hours). Bars are mean $\pm$ SD. * $p < 0.0001$.

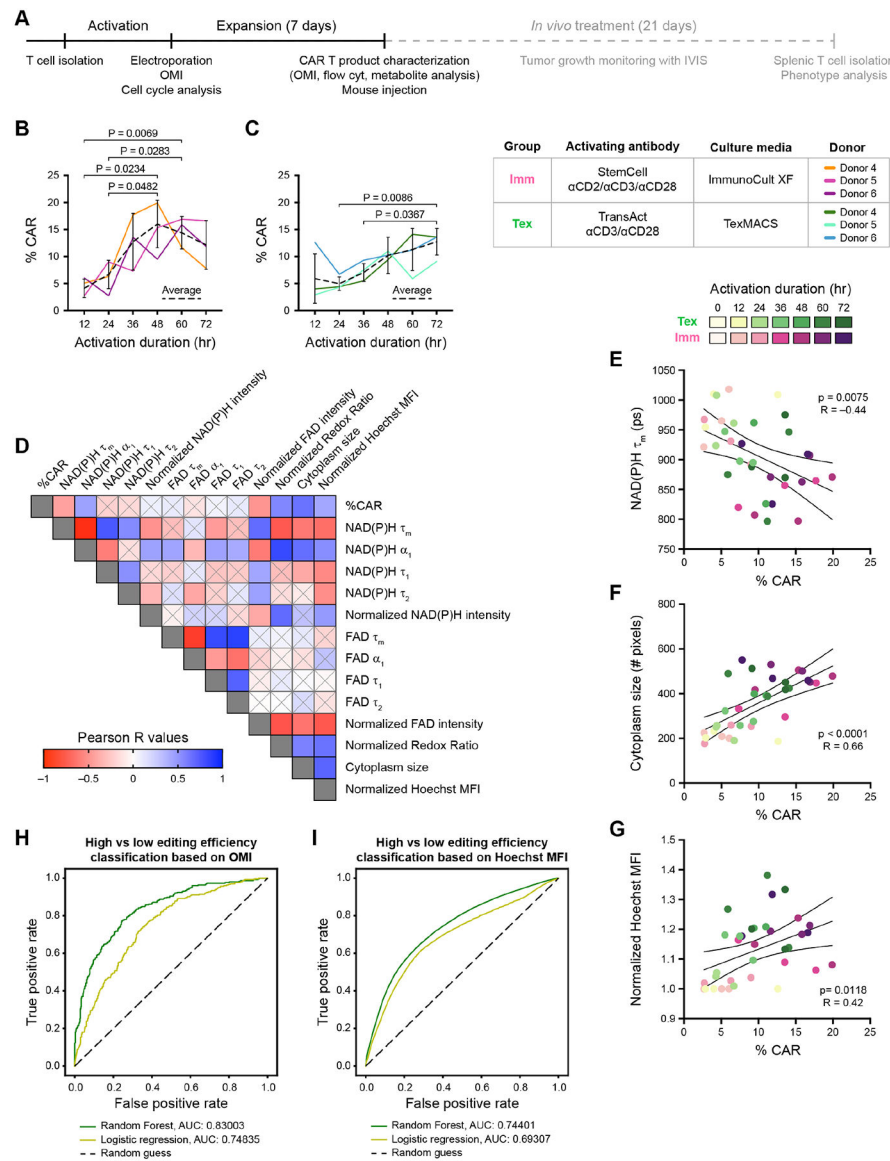


Fig. 3: OMI identifies optimal timeframe for CRISPR/Cas9 genome editing.

(A) Experimental timeline. (B, C) Genome editing efficiency (%CAR positivity) for T cells activated with (B) Imm and (C) Tex methods for 12–72 hours. $n = 4$ replicates across 3 donors for 12-hour activated group, and $n = 5$ replicates across 3 donors for 24-, 36-, 48-, 60-, and 72-hour activated groups, Brown-Forsythe and Welch ANOVA test with Dunnett T3 post hoc test. (D) Pearson R correlation matrix among OMI parameters, Hoechst median fluorescence intensity (MFI) at electroporation, and genome editing efficiency post expansion. NAD(P)H intensity, FAD intensity, redox ratio and Hoechst MFI were normalized to 12-hour-activated group averages. $n = 36$ samples across 3 donors. Crossed out cells represent non-significant correlations ($p > 0.05$). (E–G) Correlation between genome editing efficiency post-expansion and (E) NAD(P)H τ_m , (F) cytoplasm size, and (G) normalized Hoechst MFI. Each dot represents one sample average, color coded by activation method (Imm or Tex) and duration of activation (12–72 hours). (H, I) ROCs

and AUCs for classification models to predict gene editing outcomes, i.e. high efficiency (>15% CAR⁺ for Imm-activated T cells and >10% CAR⁺ for Tex-activated T cells) versus low efficiency, based on **(H)** OMI parameters and **(I)** normalized Hoechst MFIs at electroporation. $n = 3881$ cells for training, 970 cells for testing for **(H)** and $n = 611,841$ cells for training and 152,960 cells for testing for **(I)**. Bars are mean $\pm$ SD. * $p < 0.0001$.

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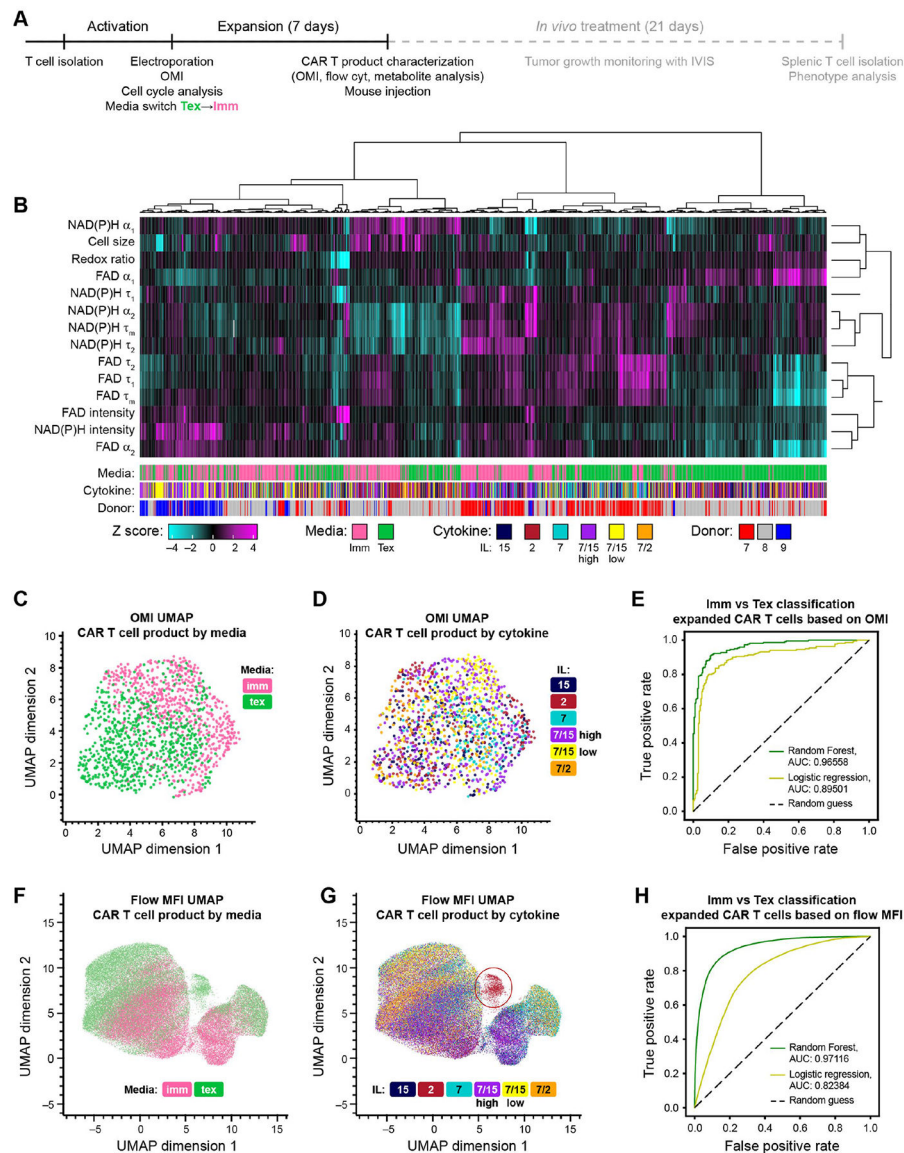


Fig. 4: Media composition has a substantial impact on CAR T cell metabolism and phenotype post-expansion.

(A) Experimental timeline. **(B)** Z-score heatmap based on 14 OMI parameters of CAR T cells (both CAR⁺ and CAR⁻) expanded in ImmunoCult XF (Imm) or TexMACS (Tex) media with various cytokines. Hierarchical clustering was determined based on Euclidean distance from population mean using Ward's criterion. **(C)** UMAP based on Euclidean distance of OMI metabolic parameters for CAR T cells at the end of manufacturing, color coded by **(C)** culture media or **(D)** supplemented cytokines. $n = 1,417$ cells from 3 donors for **(B-D)**. **(E)** ROC curves and AUCs of models to classify CAR T cells expanded in ImmunoCult XF or TexMACS media based on OMI parameters. $n = 1,134$ cells for training and $n = 283$ cells for testing. **(F-G)** UMAP based on Euclidean distance of surface marker median fluorescence intensity (MFI) (CD27, CD45RO, CD62L, CD28, CD45RA and CCR7), color coded by **(F)** culture media or **(G)** supplemented cytokines. $n = 87,317$ cells across 3 donors. **(H)** ROC curves and AUCs of RF and LR classifiers to classify T cells expanded in

ImmunoCult XF or TexMACS media based on surface marker expression. $n = 52,390$ cells for training; $n = 34,927$ cells for testing.

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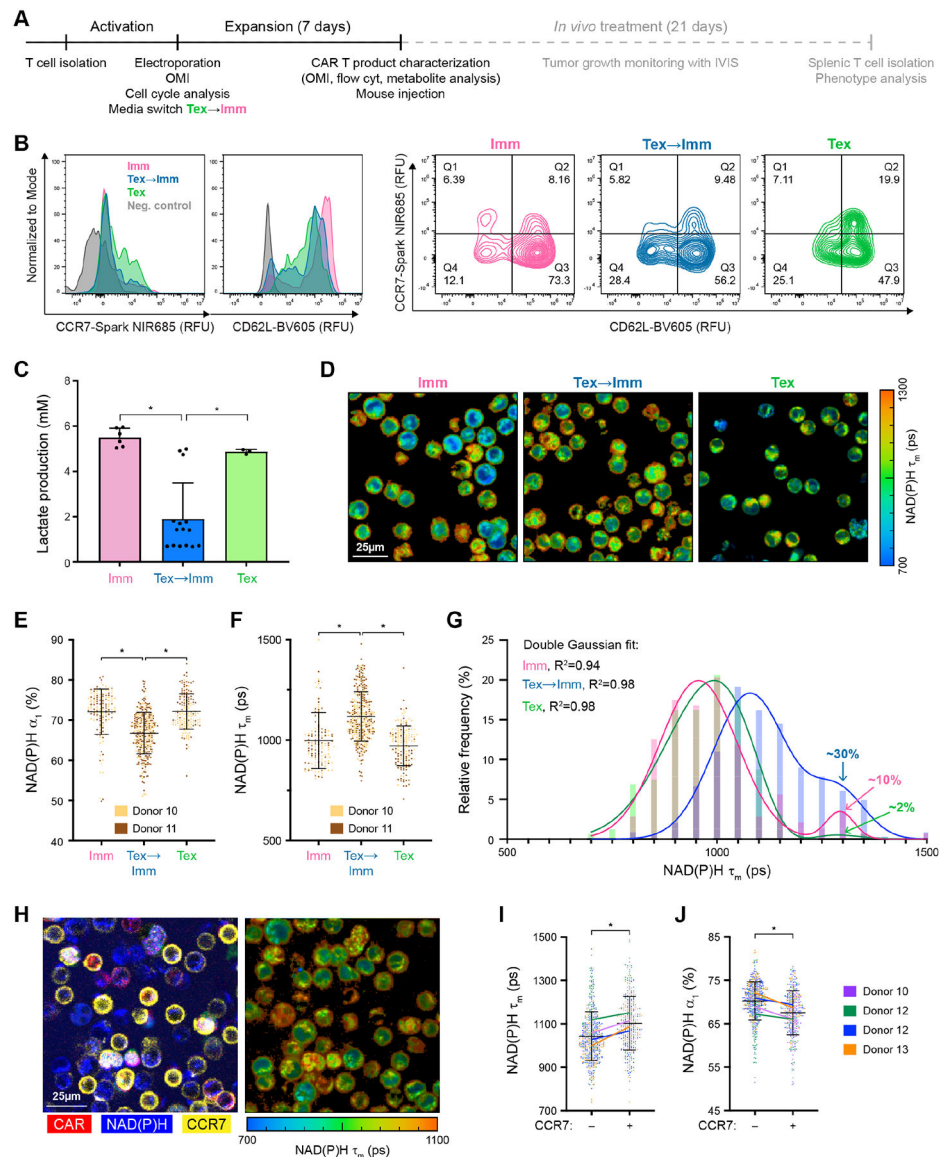


Fig. 5: CAR T cells undergoing media transition showed a distinct OMI metabolic profile with lower glycolytic activity compared to those expanded in singular media.

(A) Experimental timeline. (B) CCR7 (left) and CD62L (right) expression profile based on relative fluorescence units (RFUs) of pre-infusion CAR T cells. (C) Lactate production from 1 million CAR T cells within 24 hours. $n = 10$ Imm, 17 $\text{Tex} \rightarrow \text{Imm}$, and 5 Tex CAR T samples, Brown-Forsythe and Welch ANOVA test with Dunnett T3 test for multiple comparisons against $\text{Tex} \rightarrow \text{Imm}$ group. (D) Representative NAD(P)H τ_m images and quantification of (E) NAD(P)H α_1 and (F) NAD(P)H τ_m of CAR T cells expanded in Imm, Tex, or $\text{Tex} \rightarrow \text{Imm}$ media conditions. Brown-Forsythe and Welch ANOVA test with Games-Howell's test for multiple comparisons. (G) NAD(P)H τ_m histogram of CAR T cells post expansion. R^2 value represents goodness of fit to a double Gaussian distribution. Percentages of cells in the high NAD(P)H τ_m Gaussian are noted. (E–G) $n = 143$ Imm CAR T cells, 345 $\text{Tex} \rightarrow \text{Imm}$ CAR T cells, 160 Tex CAR T cells from 2 donors. (H) Representative immunofluorescence and NAD(P)H τ_m image and quantification of (I)

NAD(P)H τ_m and (J) NAD(P)H α_1 from CCR7⁺ and CCR7⁻ CAR T cells expanded in the Tex→Imm media condition. $n = 519$ CCR7⁻ CAR T cells and 371 CCR7⁺ CAR T cells across 4 donors, two-sided Mann-Whitney test. Bars are mean $\pm$ SD. * $p < 0.0001$.

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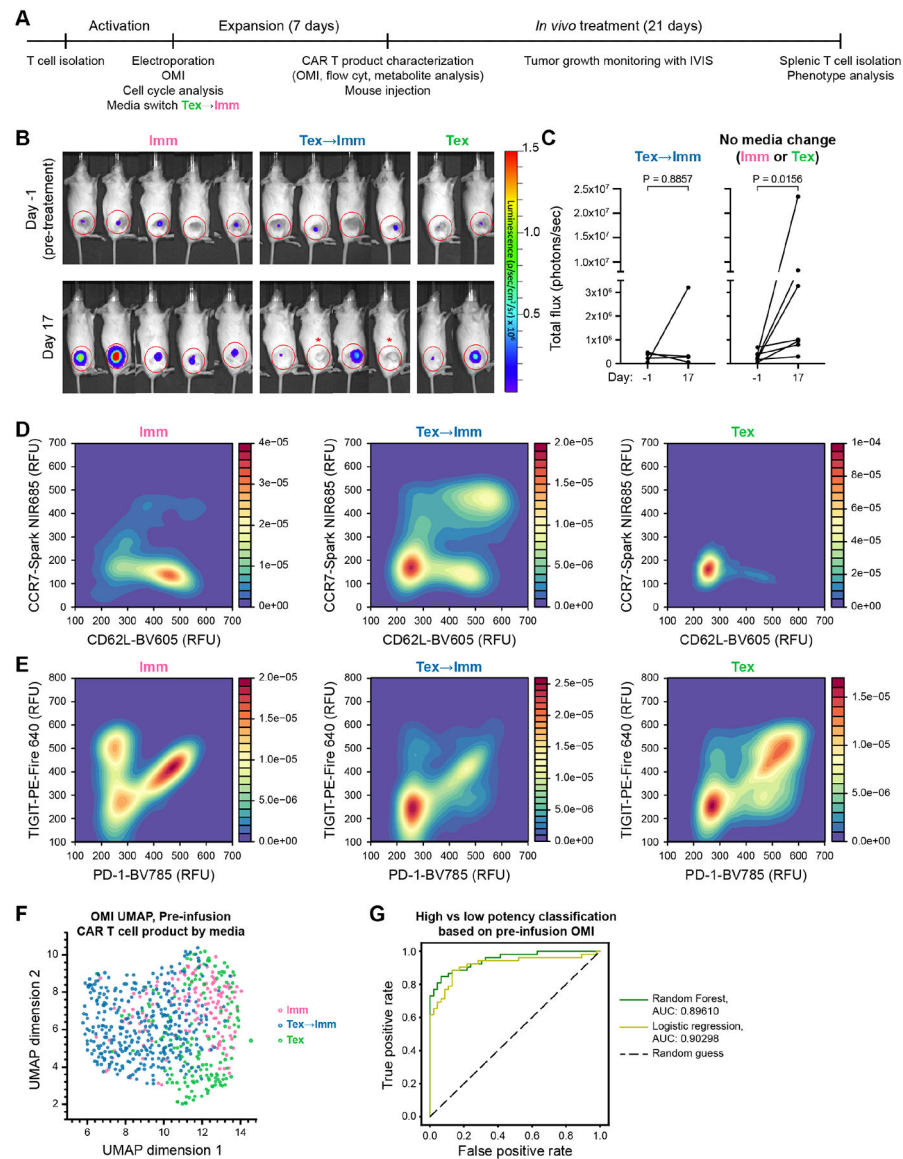


Fig. 6: OMI metabolic features of CAR T cells pre-infusion predict in vivo potency. (A) Experimental timeline. (B) Representative images of NSG mice bearing GD2+ Luciferase CHLA-20 tumors before (top row) and 17 days after (bottom row) intravenous injection with 4 million CAR⁺ T cells/mouse. Red asterisks represent tumor-free mice on day 17. (C) Tumor flux in NSG mice before (day -1) and after (day 17) treatment with Imm, Tex→Imm, and Tex CAR T cells. $n = 2-5$ mice/condition across 2 donors, paired non-parametric t-test (Wilcoxon test). (D, E) Expression of (D) stem central memory and (E) exhaustion markers in splenic T cells post treatment. 2D kernel density estimation plots based on MFIs of surface markers (D) CCR7 and CD62L and (E) PD-1 and TIGIT of CD45⁺ T cells isolated from treated mouse spleens after 21 days of CAR T treatment. $n = 9,569$ cells from 11 mice across 2 donors. (F) UMAP projection based on Euclidean distance metric of OMI parameters for CAR T cells that were activated and expanded in Imm, Tex, or Tex→Imm conditions. $n = 648$ cells from 2 donors. (G) ROC curves and corresponding

AUCs of 2 models to classify high potency (Tex→Imm) versus low potency (Imm and Tex groups) CAR T cells based on OMI parameters of pre-infusion products. $n = 518$ cells for training and $n = 130$ cells for testing. * $p < 0.0001$.

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Table 1. List of 14 OMI parameters (13 metabolic features + cytoplasm size).

OMI parameters were derived from autofluorescence intensity and fluorescence lifetimes of NAD(P)H and FAD. Each parameter was measured from the cytoplasmic region on a single-cell level using manually segmented masks.

List of OMI parameters	
NAD(P)H intensity	Number of NAD(P)H photons
NAD(P)H τ_1	Fluorescence lifetime of free NAD(P)H
NAD(P)H τ_2	Fluorescence lifetime of bound NAD(P)H
NAD(P)H α_1	Fraction of free NAD(P)H
NAD(P)H α_2	Fraction of bound NAD(P)H
NAD(P)H τ_m	Mean NAD(P)H fluorescence lifetime = $\alpha_1\tau_1 + \alpha_2\tau_2$
FAD intensity	Number of FAD photons
FAD τ_1	Fluorescence lifetime of bound FAD
FAD τ_2	Fluorescence lifetime of free FAD
FAD α_1	Fraction of bound FAD
FAD α_2	Fraction of free FAD
FAD τ_m	Mean FAD fluorescence lifetime = $\alpha_1\tau_1 + \alpha_2\tau_2$
Redox ratio	$\frac{NAD(P)H_{intensity}}{NAD(P)H_{intensity} + FAD_{intensity}}$
Cytoplasm size	Number of pixels in cytoplasm mask